



STRATHMORE UNIVERSITY

HealthLink Kenya:

A Predictive Framework for Healthcare Access and Referral Planning

*A dissertation submitted to Strathmore University in partial
fulfilment of the requirements for the degree of
Master of Science in Data Science and Analytics*

Submitted by

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Strathmore University

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*This thesis is submitted in partial fulfilment of the requirements
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at

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Letter of Approval

This thesis entitled *HealthLink Kenya: A Predictive Framework for Healthcare Access and Referral Planning*, submitted by Rutendo Julia Kandeya (ID: 168332) in partial fulfilment of the requirements for the degree of Master of Science in Data Science and Analytics at Strathmore University, has been read and approved as meeting the required standard.

We, the undersigned, certify that this work was carried out under our supervision and is suitable for submission to the School of Computing and Engineering Sciences.

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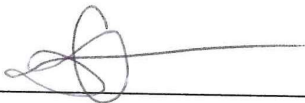
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Abstract

Kenya's public health system allocates resources reactively rather than in anticipation of demand, creating preventable access gaps among geographically remote and economically marginalised populations. This study developed *HealthLink Kenya*, a modular, two-stage predictive framework grounded in the Andersen Behavioural Model, using nationally representative microdata from the 2022 Kenya Integrated Household Health and Economic Survey (KNBS; $n = 99,031$; analytical $n = 73,059$). A nine-algorithm tournament identified XGBoost as the optimal classifier for initial care access (Stage 1: $AUC = 0.8144$, $F_1 = 0.8091$) and follow-up retention (Stage 2: $AUC = 0.8510$, $F_1 = 0.9836$), with bootstrap validation ($n = 1,000$ iterations) confirming stability across both stages. TreeSHAP analysis identified predisposing factors as the dominant predictors (73.2% of model weight), while `distance_from_facility` was the most influential individual feature (12.9%), reflecting perceived rather than verified geographic proximity a material limitation of the study's spatial conclusions. A severe 28.74:1 class imbalance constrained the baseline retention model to near-zero dropout recall (5%); SMOTE oversampling raised recall to 53.4% (precision 0.208, AUC 0.829), representing the recommended production configuration. The framework was translated into the *HealthLink Kenya* Streamlit dashboard, a dual-profile decision-support tool. Social workers access a geospatial and predictive triage view to identify health-evasive households; patients access a simplified personal interface for facility navigation and plain-language retention guidance. Analytical validation is complete; operational adoption requires structured field deployment, institutional mandate, and a pre-deployment ethical governance audit before county-level rollout.

Keywords: health-seeking behaviour, XGBoost, SHAP interpretability, Andersen Behavioural Model, healthcare access, referral planning, Kenya, predictive modelling, data-driven decision support.

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1 Introduction

1.1 Background to the Study

Healthcare accessibility remains a major global challenge, particularly in low- and middle-income countries (LMICs) where uneven facility distribution, transportation barriers, and limited health system coordination delay treatment and reduce service utilisation. In Kenya, this challenge is compounded by the persistent fragmentation of health data systems, which constrains the ability of planners and administrators to anticipate demand, coordinate referrals, and allocate resources efficiently. Most studies on health-seeking behaviour (HSB) in Kenya have remained largely descriptive and operationally isolated ([Mwaura et al., 2025](#)), limiting their integration across health departments and administrative levels. Existing digital tools for analysing HSB are often rigid, proprietary, and unsuitable for low-connectivity environments, making them difficult to scale in resource-limited settings. A further critical absence exists in interoperable frameworks that can connect behavioural insights with geospatial, clinical, and administrative data ([Musiega et al., 2023](#)).

This study addressed this gap by developing *HealthLink Kenya*, a modular, two-stage predictive framework that harmonises health-seeking behaviour data from the 2022 Kenya Integrated Household Health and Economic Survey (KNBS) and translates analytical outputs into a practical decision-support dashboard for facility-level referral planning. The framework was grounded in the Andersen Behavioural Model, which organises determinants of healthcare utilisation into predisposing, enabling, and need factors, providing a theoretically coherent structure for both feature engineering and result interpretation. Crucially, the framework was designed around two underserved actors: patients who navigate care pathways under geographic and informational constraints, and social workers who conduct community outreach without access to data-driven prioritisation tools.

1.1.1 Big Data Considerations

The rapid growth of big data and artificial intelligence (AI) in healthcare introduces both opportunities and risks for public health systems in LMICs. In Kenya, challenges including poor data quality, privacy concerns, and weak enforcement of data protection legislation risk undermining public trust in digital health tools, despite the presence of formal legal protections (Masinde et al., 2025). These conditions underscore the need for research methods that are scalable, context-sensitive, and embedded with strong data governance safeguards.

The value of flexible, modular design has been demonstrated across multiple Kenyan digital health contexts. A modular HIV PrEP training programme improved provider knowledge while accommodating scheduling constraints in busy clinical settings (Irungu et al., 2022). Similarly, blended training programmes for midwives in Kenya and Nigeria enhanced learning outcomes despite persistent gaps in digital literacy and internet reliability (Shikuku et al., 2024). A stakeholder-led framework for neglected tropical diseases further demonstrated that tools remain practical and sustainable when guidelines are developed with end-user input and updated regularly (Ng’etich et al., 2021).

Collectively, these precedents establish that frameworks designed for diverse, resource-limited populations must leverage affordable and flexible digital tools (Odeny et al., 2020), embed ethical and context-aware analytics, and maintain interpretability for non-technical end-users. These principles directly informed the design choices made in this study, particularly the emphasis on open-source tooling, modular preprocessing, and SHAP-based interpretability.

1.2 Problem Statement

Despite the widespread adoption of digital health systems in Kenya, healthcare data remain fragmented, underutilised, and poorly integrated across administrative levels. Existing platforms prioritise clinical record management and aggregate reporting rather than operational decision support, with the result that referral decisions are frequently driven by incomplete

information, informal networks, or localised knowledge rather than integrated analytical tools (Musiega et al., 2023). This fragmentation limits efficient resource allocation, creates barriers to timely treatment, and delays care for the populations most in need.

Two categories of user are left structurally unsupported by current systems. Patients navigating Kenya's public health system make access decisions under uncertainty, often without knowledge of which facilities are geographically reachable, NHIF-accredited, or appropriate for their needs. Social workers and Community Health Volunteers, tasked with reaching health-evasive households across peri-urban and rural coverage areas, operate without data-driven tools to identify or prioritise at-risk individuals. Outreach in this environment is necessarily reactive rather than predictive.

These challenges are compounded by rigid system designs that overlook regional disparities in population distribution, facility capacity, and transportation barriers (Mwaura et al., 2025), and by infrastructural constraints including unstable connectivity and low data privacy awareness that complicate the ethical use of digital health tools (Masinde et al., 2025).

To address this gap, this study developed a data-driven analytical framework integrating geographic, facility-level, and behavioural data into a modular prediction pipeline and dual-profile decision-support dashboard serving social workers through predictive risk triage and geospatial outreach mapping, and patients through personalised care pathway guidance, both grounded in empirically validated predictive models.

1.3 Research Objectives

The primary objective of this study was to develop, implement, and evaluate a modular, interpretable data-driven framework for health-seeking behaviour analytics, assessing its predictive accuracy, interpretability, and operational utility for supporting evidence-based referral planning and decision-making within Kenya's public health system.

1.3.1 Specific Objectives

- **Objective 1:** To evaluate the key structural, geographic, and socioeconomic determinants of health-seeking behaviour in LMICs and Kenya, identifying the limitations of traditional analytical approaches.
- **Objective 2:** To conduct a detailed analysis of machine learning and explainable AI (XAI) studies in healthcare, evaluating their methodological approaches, weaknesses, and capacity to address class imbalance.
- **Objective 3:** To design and implement a reproducible, Python-based data pipeline capable of harmonising heterogeneous datasets in strict alignment with the Andersen Behavioural Model.
- **Objective 4:** To develop, evaluate, and operationalise a two-stage supervised machine learning framework into an interactive decision-support dashboard (*HealthLink Kenya*), directly supporting facility-level referral planning and community outreach.

1.4 Research Questions

Research Question 1:

What structural, geographic, and socioeconomic factors most strongly determine health-seeking behaviour and provider selection within Kenya's health system, and how can interpretable models quantify their relative influence?

Research Question 2:

How can heterogeneous health-seeking behaviour datasets be harmonised into a modular, reproducible, and interpretable analytical pipeline aligned with the Andersen Behavioural Model?

Research Question 3:

To what extent can a two-stage supervised machine learning framework accurately predict initial care access and follow-up care retention among Kenyan households, and how ef-

fectively does the resulting decision-support system enhance evidence-based planning and resource allocation within Kenya's public health system?

1.5 Research Gap

Existing studies on healthcare accessibility and health-seeking behaviour have produced valuable insights into the socioeconomic and geographic factors influencing healthcare utilisation. Concurrently, digital health platforms such as electronic medical record systems and national health information infrastructures have improved data collection and management. However, these systems have largely focused on clinical documentation, aggregate reporting, or high-level policy monitoring, rather than operational decision support at the facility level (Musiega et al., 2023).

While interoperability standards such as DHIS2 integrations and FHIR-based data exchange facilitate health information sharing, they do not inherently provide analytical frameworks that transform fragmented accessibility data into actionable tools for healthcare providers. Existing architectures primarily address data exchange and reporting infrastructure; this study focused instead on the analytical transformation of health-seeking behaviour datasets into operational referral decision-support tools. In particular, limited prior research has translated heterogeneous accessibility datasets into systems capable of assisting both patients and social workers based on geographic accessibility and behavioural risk.

As a consequence, hospitals in many LMICs continue to rely on informal knowledge or incomplete information when determining appropriate referral destinations. This highlights a persistent gap between the availability of healthcare data and the development of practical analytical tools that support referral planning and operational coordination. This study addressed that gap by developing a modular data pipeline and interpretable decision-support dashboard grounded in nationally representative survey data and validated machine learning models.(Moucheraud et al., 2022)

1.6 Research Justification

The proposed framework was grounded in principles of modularity, adaptability, and interpretability to address the operational challenges associated with fragmented healthcare accessibility data. Many digital health systems in LMICs remain difficult to scale or adapt due to rigid architectures and limited analytical integration. Modular data architectures offer a practical alternative by enabling incremental development, local customisation, and reusable analytical components that can be adapted across different health system contexts (Odeny et al., 2020).

To support this approach, the study implemented a Python-based end-to-end data pipeline that harmonised the four feature modules of the Andersen Behavioural Model health utilisation behaviour, predisposing socio-demographic factors, enabling socio-economic factors, and geographic access factors into a reproducible analytical structure. By standardising data preparation, transformation, and modelling processes, the pipeline promoted transparency, repeatability, and extensibility across health system contexts.

The analytical layer of the framework emphasised methods that balance interpretability with predictive capability. XGBoost was selected as the optimal classifier following a nine-algorithm tournament evaluated on ROC-AUC, given the class-imbalanced nature of the dataset. TreeSHAP visualisations ensured that model outputs remained transparent and understandable for health administrators and policymakers who rely on these insights for operational planning.

The framework's operational utility, however, is contingent on more than technical performance. Social workers and Community Health Volunteers in Kenya's public health system operate under significant structural constraints high caseloads, limited digital literacy, unpredictable connectivity, and in many cases voluntary or low-compensated roles with high turnover rates (Njiraini and Hussein, 2020). A dashboard that is analytically sound but requires substantial training, uninterrupted data access, or significant time investment per case risks becoming operationally unused regardless of its predictive performance.

This study explicitly distinguishes between analytical validation and operational adoption as two separate achievements. The former is demonstrated through the modelling and testing work reported in Chapters 3–5. The latter requires structured usability trials, institutional mandate, and capacity-building investment that fall outside the scope of this research prototype, and are identified as the primary priority in the future work programme (Section 7.3).

The framework ultimately integrated its analytical outputs into the *HealthLink Kenya* Streamlit dashboard, which translates complex predictive outputs into two distinct operational interfaces. Social workers access a full-profile geospatial and predictive triage view, enabling them to identify health-evasive households and plan targeted community outreach. Patients access a simplified, light-themed personal interface, enabling self-directed identification of accessible care facilities based on their individual enabling and geographic constraints.

1.7 Research Scope

This study focused on developing an analytical framework for evaluating healthcare accessibility and supporting referral planning within Kenya’s healthcare system. The framework integrated secondary data from the KNBS 2022 Health-Seeking Behaviour Survey together with facility-level information to analyse structural factors influencing healthcare accessibility.

The analysis examined key determinants including education level, wealth index, distance to care, insurance status, and residential area, as well as provider categories such as clinics, county hospitals, national referral hospitals, and traditional healers. These variables were organised into the four modules of the Andersen Behavioural Model and used to construct analytical indicators informing the identification of accessible healthcare providers and potential referral options.

The study was bounded by several scope constraints. The analysis relied on secondary survey data rather than clinical medical records, and the dataset did not include high-resolution

spatial coordinates that would enable detailed geospatial modelling at the individual facility level. The framework focused on accessibility and referral support rather than clinical diagnosis prediction or real-time hospital capacity monitoring. The retention model (Stage 2) was further constrained by a severe class imbalance (96.2% retained), which limited its utility for identifying true dropout cases despite a strong overall AUC.

1.8 Research Limitations

Generalisability and Local Context. Although the framework was designed for broad applicability, local cultural, administrative, and infrastructural factors require contextual adaptation for deployment in specific regions. The absence of highly granular sub-county variables may have constrained predictive accuracy for isolated sub-populations.

Data Availability and Quality. The dataset’s heavy skew towards positive healthcare utilisation (82% visited, 96% retained) introduced a sensitivity bias in both models, prioritising the identification of access over exclusion. Missing and inconsistent values required median imputation for continuous variables and mode imputation for categorical variables. Additionally, the imputation of missing target variable values (`visited_provider` and `revisited_provider`) as positive responses introduces a conservative analytical assumption that is acknowledged as a limitation of the study design.

Absence of High-Resolution Spatial Coordinates. The KNBS 2022 dataset provides self-reported estimates of distance to the nearest facility rather than GPS-derived geodesic measurements. This distinction is analytically consequential: self-reported distance is subject to respondent perception bias, does not account for road network topology or seasonal accessibility, and cannot support travel-time catchment modelling at the sub-county level. All spatial analysis in this study is therefore bounded by the precision of survey-reported distance, and the distance-decay thresholds identified by the GAM should be interpreted as approximations rather than precise geometric boundaries.

This limitation was empirically validated as part of the study’s future work scaffold. A Nominatim-based geocoding pipeline was applied to the 10,668 unique facility names in the

KNBS dataset, successfully geocoding 1,558 facilities (14.6%). Geodesic distances were then computed from each geocoded facility's verified GPS coordinates to the centroid of its inferred county, producing a geodesic distance estimate for 23,259 analytical rows (23.5% coverage). The Pearson correlation between self-reported and geodesic distance across these matched rows was $r = -0.040$ a near-zero coefficient indicating that respondents' perceptions of distance bear no systematic linear relationship to verified geographic distance. This finding constitutes a material limitation of the study's spatial conclusions: the GAM-derived 35 km threshold and the SHAP distance attribution should be interpreted as reflecting respondent perceptions, not verified geographic access constraints. The result empirically substantiates the theoretical concern about perception bias and suggests that replacing self-reported distance with GPS-derived geodesic measurements would introduce materially different and more analytically valid spatial signal into the model.

Cross-Sectional Design and Absence of Longitudinal Linkage. The KNBS 2022 survey is a single cross-sectional instrument. Individual respondents cannot be linked across survey waves, preventing the construction of care-seeking trajectories over time. As a result, the retention model captures a static snapshot of revisit behaviour rather than a dynamic view of long-term engagement with the health system. Patients who cycle in and out of care, switch providers between survey rounds, or whose access barriers change with life circumstances are analytically invisible in this design. Individual-level linkage across waves would require either a panel survey component or probabilistic record linkage to national health information registries, both of which fall outside the scope of secondary data analysis ([Masinde et al., 2025](#)).

Privacy and Ethical Constraints. The use of anonymised secondary data protected participant confidentiality but resulted in the loss of contextual depth that primary data collection would have provided. Facility-level quality variables including medicine stockout rates, clinician-to-patient ratios, and patient-reported waiting times are absent from the KNBS instrument, constraining the model's ability to distinguish geographic access failure from quality-of-care failure at the prediction stage rather than retrospectively through error analysis.

Practical and Technical Constraints. The modular pipeline requires Python-based computational infrastructure and technical expertise to execute and maintain. Long-term adoption will depend on robust documentation, local capacity-building, and institutional support for sustainable deployment.

Interpretability versus Complexity. While interpretable models promoted transparency and stakeholder trust, ensemble methods such as XGBoost may compress predictive precision relative to deep learning alternatives. Even with TreeSHAP visualisations, unmeasured or context-specific variables for example facility opening hours, seasonal road access, or household illness severity may remain undetected by the model.

These limitations underscore the importance of local adaptation, transparent documentation, and realistic expectations regarding model accuracy in operational deployment. Future iterations of this pipeline should expand regional customisation, improve data harmonisation techniques, and strengthen capacity-building for locally managed use.

1.9 Utilisation and Dissemination of Findings

To ensure that the study’s findings translated from theoretical analysis to practical application, the outputs and developed models were structured for utilisation and dissemination through three targeted channels.

First, for **operational utilisation**, the core findings and predictive models were translated into the *HealthLink Kenya* dashboard, deployed via Streamlit as an interactive decision-support interface. The dashboard enables health planners to simulate access gap scenarios, explore the structural determinants of non-utilisation, and identify facilities for targeted referral based on model-derived accessibility scores. It further serves as a direct empowerment tool for the two primary user groups: social workers who access geospatial and predictive triage outputs, and patients who receive personalised facility guidance through a simplified self-navigation interface.

Second, to support **policy dissemination**, the study’s findings were structured to inform evidence-based recommendations for Kenya’s Ministry of Health, particularly regarding

the distance-decay thresholds identified through the Generalised Additive Model (GAM) analysis and their implications for facility placement standards and mobile clinic deployment.

Third, to facilitate **open-access research utilisation**, the complete modular code pipeline was hosted on a public repository. By providing data cleaning scripts, serialised model artefacts, and metadata documentation, the framework complies with FAIR data principles (Findable, Accessible, Interoperable, Reusable) and allows other researchers to replicate the study or adapt the modules for different health domains and geographic contexts.

In summary, this chapter established the background, problem statement, research objectives, and scope guiding the study, consistently motivated by the need for a modular, interpretable, and context-aware analytical framework serving two underserved actors in Kenya's health system: the patient navigating care under constraint, and the social worker conducting outreach without data-driven support. The next chapter reviews existing literature on health-seeking behaviour, data interoperability, and analytical modelling approaches, identifying conceptual and methodological gaps that further justify the proposed framework and situating it within current research trends in digital health analytics and public health decision support.

2 Literature Review

2.1 Introduction

This chapter critically examines the theoretical, methodological, and operational literature relevant to healthcare accessibility analytics in low- and middle-income country (LMIC) contexts. The review is organised around five interconnected themes, each of which contributes to the analytical and design rationale of the *HealthLink Kenya* framework.

Three substantive gaps motivate the study and run as a thread through each section. First, Andersen’s Behavioural Model (Li et al., 2023; Solanke et al., 2023b) the dominant conceptual framework for healthcare utilisation research is widely applied descriptively in LMIC settings but rarely translated as a structured predictive feature engineering architecture. Second, physical distance to care is routinely modelled as a linear predictor, suppressing nonlinear threshold effects that carry direct policy consequences (Waqas et al., 2024; Belay et al., 2025). Third, despite demonstrable advances in machine learning and explainable artificial intelligence (XAI), the deployment of interpretable predictive models within African health information systems remains practically underdeveloped (Chepkirui et al., 2025; Masinde et al., 2025).

Woven across all three gaps is a consistent finding from the health-seeking behaviour literature: the populations most at risk of exclusion from formal healthcare health-evasive individuals and under-resourced social outreach workers are also the populations least served by existing digital health tools. The *HealthLink Kenya* framework responds directly to this dual deficit, designing separate but analytically unified interfaces for the patient (P) navigating care under constraint, and the social worker (SW) conducting community outreach without data-driven prioritisation support.

The chapter is structured as follows. Section 2.2 reviews Andersen’s Behavioural Model and the case for its computational operationalisation. Section 2.3 examines the distance-decay literature and the methodological limitations of linear spatial modelling. Section 2.4 reviews ensemble machine learning methods and explainable AI in the context

of health-seeking behaviour prediction. Section 2.5 examines the evidence on health-evasive populations, community health worker outreach, and patient self-navigation and directly addresses the ethical implications of deploying predictive risk tools in these contexts. Section 2.6 covers data harmonisation and deployment architecture. Section 2.7 synthesises the five gaps addressed by this study, and Section 2.9 presents the organising conceptual framework.

2.2 Theoretical Foundation: Andersen’s Behavioural Model

2.2.1 Origins and Core Structure

Andersen’s Behavioural Model of Health Services Use, first proposed in the 1960s and subsequently refined (Andersen, 2024; Li et al., 2023), organises the determinants of healthcare utilisation into three analytically distinct categories: *predisposing characteristics* demographic and social attributes that precede illness, including age, gender, and education level; *enabling resources* structural conditions that facilitate or impede access, including income, insurance coverage, and proximity to facilities; and *perceived or evaluated need* subjective and clinical illness assessments that motivate care-seeking. The model’s value lies in providing a taxonomically coherent classification system that enables comparative research on healthcare disparities across diverse institutional and geographic contexts.

The model has been applied across a wide range of LMIC health contexts. Orhan and Kurutkan (2025) applied Andersen’s three factor categories directly to a machine learning pipeline for predicting total healthcare demand, demonstrating that the taxonomy can serve as both a descriptive organiser and a feature engineering scaffold. Kassaw et al. (2025) similarly operationalised the model to predict healthcare access among rural pregnant women in Ethiopia, confirming that predisposing and enabling factors jointly dominate access outcomes in resource-limited settings a finding that directly anticipates the *Andersen Reversal* documented in this study’s SHAP analysis.

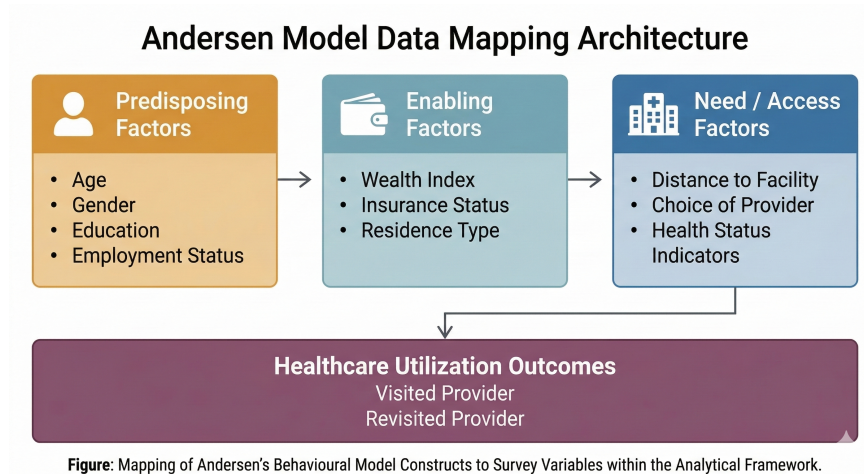


Figure 2.1: *Andersen's Behavioural Model: policy application framework illustrating the hierarchical relationship between predisposing, enabling, and need factors as determinants of healthcare utilisation.*

2.2.2 Limitations in LMIC Contexts

Andersen's original formulation was constructed within high-income healthcare systems characterised by stable institutional infrastructures, consistent data standards, and limited geographic heterogeneity. Three structural limitations constrain its direct application in LMIC contexts.

First, the model treats enabling factors as static attributes, whereas in environments such as Kenya, insurance coverage, facility availability, and transport access vary substantially across seasons, administrative boundaries, and policy cycles (Nyangena et al., 2021; Chepkirui et al., 2025). Second, fragmented digital health infrastructures and inconsistent coding standards prevent the integration of behavioural, spatial, and administrative datasets into unified analytical structures (Masinde et al., 2025), rendering the model descriptive rather than practically applicable. Third, empirical applications in sub-Saharan African contexts consistently find that enabling factors particularly geographic distance and economic enabling conditions dominate utilisation decisions to a degree that the original formulation does not adequately weight (Solanke et al., 2023a; Orhan and Kurutkan, 2025).

Rather than invalidating the model, these constraints highlight the necessity for its computational adaptation. The current study addresses this by translating Andersen's three

factor categories directly into four structured feature engineering modules, transforming the framework from a descriptive taxonomy into a predictive architecture.

2.2.3 Implementation Through Machine Learning

Recent methodological developments in machine learning and explainable AI open substantive pathways for modernising Andersen’s framework beyond descriptive application. SHAP attribution methods enable the estimation of nonlinear, high-order relationships among predisposing, enabling, and need-related features while preserving interpretability accessible to non-technical policymakers (Orsini et al., 2022; Yehuala, 2024). Emerging semantics-aware data harmonisation standards further support the integration of heterogeneous datasets from multiple administrative and geographic domains (Dylag et al., 2024; Saras et al., 2025).

Building on these precedents, this study translates the Andersen taxonomy as a structured feature engineering pipeline, anchoring the study’s primary methodological contribution and providing the theoretical basis for interpreting SHAP outputs in terms of the model’s original factor categories.

2.3 Distance and Spatial Barriers to Healthcare Access

2.3.1 The Distance-Decay Effect

Physical distance to healthcare facilities is consistently identified as the most influential enabling factor in Andersen’s model across LMIC contexts (Masinde et al., 2025; Ng’etich et al., 2021). Rural and geographically marginalised communities face compounding access barriers: long travel distances are associated with inadequate transportation infrastructure, higher indirect costs, and diminished insurance penetration, all of which independently suppress both initial utilisation and treatment adherence (Belay et al., 2025). The spatial distribution of health facilities in Kenya exemplifies the severity of this challenge: urban counties particularly Nairobi and Central Rift concentrate the majority of private and specialised facilities, while arid and semi-arid counties in the north and north-east face

acute service shortfalls, with large populations residing beyond any reasonable travel radius (Moturi et al., 2022).

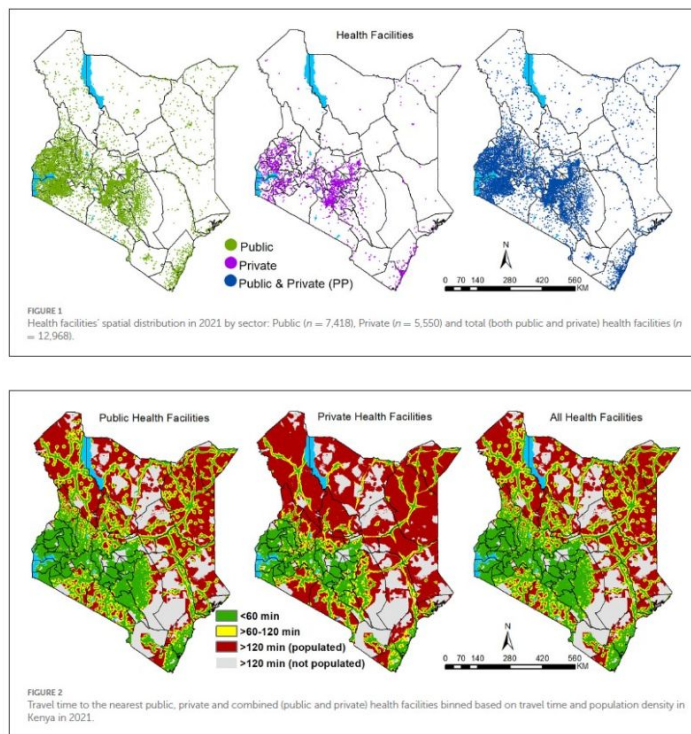


Figure 2.2: *Spatial distribution of health facilities across Kenya by sector and travel-time catchment. Public facilities remain concentrated in urban centres, while large proportions of the rural north and north-east fall outside a 60-minute travel radius, illustrating the structural geographic access inequalities documented in the literature (Moturi et al., 2022).*

Geospatial analyses corroborate that residents of remote counties regularly travel in excess of two hours to reach formal healthcare services (Belay et al., 2025). Extended travel burdens do not merely reduce care-seeking frequency: they compound longitudinally, eroding adherence to follow-up appointments, medication schedules, and long-term treatment engagement outcomes directly relevant to the retention modelling component of this study. For the social worker (SW) operating across large peri-urban or rural coverage areas, the 35 km GAM-derived threshold operationalised in *HealthLink Kenya* is subsequently **applied as the grid zone boundary** for outreach prioritisation.

2.3.2 The Inadequacy of Linear Distance Modelling

The methodological treatment of distance as a linear predictor constitutes a systematic error in the existing literature. Linear distance assumptions understate the complexity of the access environment in two specific ways. First, the relationship between distance and utilisation exhibits pronounced threshold effects: utilisation remains relatively stable up to a critical distance, beyond which it declines sharply (Waqas et al., 2024). Imposing linearity suppresses the identification of these thresholds and produces misspecified models that underestimate exclusion risk for populations residing marginally beyond a tipping point. Second, linear coefficients cannot distinguish between populations facing infrastructural access failure (no road access) and those facing cost-based access failure (road exists but transport is unaffordable) distinctions with materially different policy implications (Belay et al., 2025).

This methodological limitation carries a direct consequence for the **P** experience. A patient whose estimated distance falls just below a linearly derived threshold is treated as equivalent to one living within easy reach; a nonlinear model reveals this as a false equivalence. The *HealthLink Kenya* framework addresses this by fitting a GAM to characterise the true functional form of the distance-utilisation relationship before applying XGBoost to capture higher-order spatial interactions.

2.3.3 Generalized Additive Models as a Methodological Advance

Generalized Additive Models (GAMs) replace fixed linear coefficients with flexible smooth spline functions, enabling the empirical estimation of nonlinear distance-utilisation relationships and the identification of policy-relevant inflection points that linear specifications cannot recover (Waqas et al., 2024; Belay et al., 2025). GAMs thus provide a methodologically rigorous basis for recalibrating facility placement standards, ambulance dispatch radii, and mobile clinic deployment schedules. In this study, the GAM serves as a complement to the XGBoost classification model: it characterises the spatial form of the distance-utilisation

relationship and derives the 35 km threshold that is subsequently operationalised as the grid zone boundary in the Nakuru County pilot deployment.

2.4 Machine Learning for Health-Seeking Behaviour Analytics

2.4.1 Ensemble Methods and Their Analytical Advantages

Ensemble learning methods have demonstrated systematic superiority over single-algorithm approaches in healthcare analytics tasks, specifically because health-seeking behaviour data exhibit high-dimensional feature interactions, nonlinear covariate relationships, and pronounced class imbalance that regression-based methods cannot adequately model (Faye et al., 2025; Taye et al., 2025). Random Forest algorithms address overfitting through bootstrap aggregation across decorrelated decision trees (Avinash et al., 2025). Gradient boosting methods including XGBoost (Wiens et al., 2025) extend this framework by sequentially training learners on residual errors, achieving higher discriminative performance in imbalanced classification tasks and demonstrating robustness on structured tabular health survey data (Soladoye et al., 2025). Seto et al. (2022) further demonstrated that gradient boosting consistently outperforms logistic regression in predicting probability for chronic disease outcomes using large-scale survey data, a finding that directly informs the algorithm tournament implemented in this study.

The nine-algorithm tournament implemented here evaluated on ROC-AUC given the class-imbalanced target distribution provides a systematic empirical basis for algorithm selection rather than relying on prior assumption, and generates the performance comparison that justifies XGBoost as the foundation of both the access (Stage 1) and retention (Stage 2) classifiers.

2.4.2 Class Imbalance as a Methodological and Policy Problem

Class imbalance is a structural property of national health survey data, not an incidental data quality issue. In populations where the majority of respondents report having accessed care, algorithms that minimise aggregate classification error will disproportionately optimise for

the majority class, producing high apparent accuracy while systematically failing to identify the minority populations most at risk of exclusion (Carvalho et al., 2025; Sani et al., 2025). In the healthcare access context, false negatives individuals predicted to access care who do not represent missed opportunities for targeted intervention: the very populations that decision-support tools are intended to serve.

This distinction carries direct operational weight for both **P** and **SW**. A patient who is misclassified as low-risk will not appear in the social worker’s outreach queue, even if they are a genuine dropout risk. A social worker operating on flawed triage outputs will systematically under-reach the most marginalised households. Accordingly, this study incorporates the `scale_pos_weight` parameter in XGBoost and evaluates model performance using F1-score and ROC-AUC rather than standard accuracy (Carvalho et al., 2025; Sani et al., 2025). The choice between SMOTE oversampling and cost-sensitive weighting and the implications of that choice for geographic coordinate integrity is documented in detail in Chapter 4.

2.4.3 Explainable Artificial Intelligence and Algorithmic Accountability

High-performance ensemble models in healthcare applications face a fundamental accountability problem: predictions that cannot be explained cannot be acted upon responsibly by clinicians, administrators, or policymakers (Alharthi et al., 2025; Awe et al., 2025). SHAP (SHapley Additive exPlanations), grounded in cooperative game theory, addresses this problem by computing the marginal contribution of each input feature to a model’s prediction, averaged across all possible feature subsets, yielding consistent, theoretically principled attribution scores (Alharthi et al., 2025). TreeSHAP (Lundberg et al., 2020) extends this to tree-based ensembles with polynomial-time computation, enabling global feature attribution across the full test set and local explanation of individual predictions.

The accountability dimension of XAI is not merely technical. For **SW** using a predictive triage tool, a SHAP output that identifies distance as the dominant driver of exclusion for one cluster and low wealth as the dominant driver for another directly informs *how* to structure the outreach whether to address transport barriers or economic ones. For **P**, a plain-language

rendering of the same SHAP output enables personalised, self-directed care navigation without requiring statistical literacy. By applying TreeSHAP to the XGBoost access model, this thesis quantifies the predictive weight of enabling factors distance, wealth index, and insurance status thereby empirically validating the Andersen model hierarchy and equipping policymakers with a prioritised account of actionable barriers (Noor et al., 2025; Kita et al., 2025).

2.5 Health-Evasive Populations, Social Outreach, Patient Self-Navigation, and Ethical Constraints

This section deliberately integrates four themes that are frequently treated in isolation in the literature: the characteristics of health-evasive populations, the evidence base for community health worker outreach, the conditions for effective patient self-navigation, and the ethical obligations generated when predictive risk tools are deployed in community settings. These themes are inseparable in practice: the design of a tool for **P** and **SW** cannot be decoupled from an honest account of who these users are, what structural constraints they face, and what harm may result from misclassification or misuse.

2.5.1 The Health-Evasive Population: Definition and Global Evidence

A structurally distinct population sits at the intersection of every major barrier documented in the health-seeking behaviour literature: individuals who do not present to formal health services not because care is entirely inaccessible, but because the cumulative weight of social, cultural, economic, and experiential deterrents exceeds their perceived benefit of engagement. This phenomenon variously described as healthcare avoidance, passive non-utilisation, or voluntary disengagement is empirically distinguishable from access failure caused by geographic exclusion (Kraschenin et al., 2008). Where geographic exclusion represents a hard supply-side constraint, health-evasive behaviour represents a demand-side failure that persists even when facilities are within reach.

Globally, health-evasive populations are disproportionately represented among specific demographic clusters: young adult males who associate care-seeking with vulnerability (Gilbert et al., 2016); indigenous and marginalised ethnic communities whose historical experiences of institutional discrimination have eroded systemic trust (Srugo et al., 2023); undocumented or informally employed individuals who fear administrative consequences of formal health registration (DeRose et al., 2015); and chronically poor households for whom the indirect costs of care transport, lost wages, childcare constitute a stronger deterrent than the direct cost of treatment itself (ASPE, 2015). What unites these groups is not the absence of a nearby facility but the presence of a barrier that no map can measure.

In Kenya, this pattern acquires a specific geographic texture. Even in counties where the urban proximity paradox holds where populations residing within a moderate distance of a facility systematically fail to utilise it evasion rather than distance is often the primary mechanism of exclusion (Noor et al., 2022). Low health literacy, distrust of biomedical paradigms, reliance on traditional healers as first-contact providers, and gender norms governing who may seek care without stigma all contribute to this pattern (Cheruto, 2020; Ofori et al., 2025). The operational consequence is that analytical frameworks designed exclusively around geographic accessibility systematically misclassify a large segment of the at-risk population: individuals who appear accessible by distance metrics but remain effectively excluded by behavioural and social determinants.

This characterisation carries a direct implication for the tool design. A patient who avoids formal health systems by disposition is by definition among the least likely users of a digital self-navigation tool requiring login, data entry, and interpretation of algorithmic outputs. This structural paradox faces all patient-facing digital health tools targeting hard-to-reach populations (Labrique et al., 2017; Valdivia et al., 2019). The design resolution adopted in this study is to reframe the patient portal's primary function: not as a tool that reverses evasion through self-directed engagement, but as a retention and re-engagement support tool most powerfully used during a mediated encounter initiated by a social worker or a **Community Health Volunteer (CHV)**). Evasion reversal is the **SW** console's function. The patient portal's function is to reduce dropout once engagement has begun.

2.5.2 Community Health Workers and Social Workers as Demand-Side Bridges

The global evidence base for **Community Health Workers (CHWs)** and social worker programme and social worker programmes consistently demonstrates that outreach-based, relationship-mediated models of care engagement produce measurable improvements in utilisation among health-evasive populations where facility-based interventions have failed (Perkins et al., 2019). CHWs operating within the communities they serve carry a form of social capital that institutional facilities cannot replicate: familiarity, linguistic and cultural fluency, and the trust accumulated through sustained personal presence. In sub-Saharan African contexts, CHW programmes have demonstrated impact across maternal and child health uptake, HIV treatment adherence, tuberculosis case-finding, and non-communicable disease screening precisely the domains in which passive, facility-dependent systems consistently underperform among marginalised populations (Perkins et al., 2019; Solanke et al., 2023b).

The critical constraint on CHW and social worker effectiveness is targeting. When outreach workers must identify at-risk or evasive households through informal community knowledge alone, their reach is bounded by personal networks and geographic familiarity. Systematic under-coverage of the most isolated or socially marginalised households those least likely to be known to anyone with institutional contact is a documented limitation of non-data-guided outreach programmes (Nelson and Madiba). A SW equipped with a risk-stratified geospatial map of their catchment area can direct limited outreach time toward the individuals least likely to self-present. The SHAP attribution layer further enriches this capability: understanding *why* a specific community cluster is predicted to be evasive enables the worker to tailor their engagement approach rather than delivering a generic intervention to a heterogeneous population.

Kenya's community health system provides an existing institutional scaffold for this model. The CHV network already deploys community-level workers tasked with first-contact health promotion and referral linkage (Njiraini and Hussein, 2020). What this infrastructure has historically lacked is a data layer: a structured, interpretable analytical tool that translates

national survey evidence and predictive model outputs into spatially actionable outreach intelligence.

Designing a tool for this population, however, is not equivalent to integrating it into practice. The CHV workforce in Kenya is characterised by high turnover, limited formal training in data interpretation, and working conditions often unpaid or incentive-based that constrain the time available for any activity beyond direct household contact (Njiraini and Hussein, 2020). Evidence from analogous data-driven outreach tools in sub-Saharan Africa consistently identifies the same implementation barriers: inadequate training time, absence of supervisory reinforcement, and the perception among frontline workers that digital tools add administrative burden without reducing physical workload (Chepkirui et al., 2025; Lilan et al., 2025). For *HealthLink Kenya* to fulfil its intended function, deployment must be accompanied by a structured onboarding protocol, a plain-language field guide that does not require prior statistical literacy, supervisory-level adoption mandates from county health management, and a feedback mechanism through which social workers can flag erroneous predictions. Without these conditions, the tool addresses a targeting gap in principle while the gap persists in practice.

2.5.3 Patient Self-Navigation: Evidence and Design Conditions

Parallel to the evidence for outreach-based models, a growing body of literature documents the conditions under which patients can effectively navigate health systems when provided with appropriately designed informational tools (Valdivia et al., 2019). Patient self-navigation the capacity of an individual to identify accessible care options, understand eligibility constraints, and make an informed facility selection without requiring clinical intermediation reduces dependence on institutional gatekeeping and has been associated with improved treatment initiation rates (Valdivia et al., 2019).

The design conditions for effective patient-facing health tools in LMIC settings are well characterised. Interfaces must minimise cognitive load, present only information directly relevant to the individual's decision context, and be optimised for daytime outdoor readability on low-cost devices with inconsistent connectivity (Labrique et al., 2017). Personalisation

presenting a user with outputs derived from their own profile rather than population averages significantly improves engagement (Aguilera et al., 2026).

These imperatives directly shaped the patient-facing profile of the *HealthLink Kenya* dashboard. The light-themed interface (#F7F9FC base, navy accent) was selected for readability in high-ambient-light field conditions. The pin-drop facility locator restricts visible outputs to facilities within a user-defined geographic grid. Asymmetric access controls ensure that **P** sees only their own data, preserving privacy while maintaining full analytical depth for authorised outreach personnel.

The evidence reviewed above, however, applies primarily to patients who have already initiated engagement with the health system. The conditions for effective self-navigation identified by Valdivia et al. (2019) awareness of the tool, motivation to use it, and sufficient health literacy to act on outputs are precisely the conditions that health-evasive populations lack by definition. Applying self-navigation evidence uncritically to a health-evasive target population would substantially overstate the patient portal's reach. The *HealthLink Kenya* patient portal is therefore most accurately characterised as a retention tool for the recently re-engaged, and a guided navigation aid for patients introduced to the system through CHV-mediated encounters, rather than a standalone self-activation tool for the persistently evasive.

2.5.4 Ethical Obligations in Predictive Outreach Systems

The deployment of predictive risk-scoring systems in community health outreach generates three categories of ethical obligation that must be treated as design constraints rather than post-hoc compliance measures. These obligations are introduced here because they directly shaped the *HealthLink Kenya* design and because they are most legible when understood against the evidence on health-evasive populations and CHV practice reviewed above.

The harm of misclassification. A predictive model operating in a triage context produces two categories of consequential error. False negatives households predicted low-risk who are genuinely high-risk result in missed outreach and potentially preventable harm. In a population already characterised by structural exclusion, this error compounds exist-

ing disadvantage: the patients most likely to be misclassified are those whose profiles are underrepresented in training data, including the rural and low-wealth households disproportionately absent from this study’s analytical dataset. False positives households predicted high-risk who do not require priority intervention result in misallocated CHV time and, more seriously, the risk of unsolicited contact that a health-evasive household may experience as intrusive, potentially reinforcing avoidance behaviour rather than reversing it (Kraschenin et al., 2008). The retention model’s operating characteristic under the SMOTE configuration approximately four false positives for every true dropout identified must be disclosed to every SW user before deployment. Aggregate performance metrics such as AUC and F1-score do not capture these real-world consequences.

Protection from risk-label stigma. A household assigned a “high-risk” classification in a community outreach context carries that label into a social environment where it may become visible to neighbours, community leaders, or family members. In Kenya’s community health system, where CHVs are embedded members of the communities they serve, the boundary between professional and social knowledge is porous. A risk score that functions as a triage tool in a clinical context can function as a social label in a community context, with consequences for dignity, standing, and the patient’s subsequent willingness to engage with the health system (Cheruto, 2020). *HealthLink Kenya’s* design response presenting risk outputs as care pathway guidance rather than risk classifications, and restricting score visibility to authenticated accounts partially mitigates this risk. Full mitigation requires community-level communication about how the system works and what a CHV visit prompted by a model recommendation means.

Algorithmic accountability. When a SW prioritises one household over another because a model recommended it, and the deprioritised household subsequently experiences a preventable health crisis, the question of accountability who decided, on what basis, and with what oversight becomes institutional rather than technical. This study addresses this obligation through TreeSHAP transparency, open-source publication of the pipeline, and the design principle that the model is a decision-support aid, not a decision-maker (Mackin et al., 2025). These are necessary conditions for accountability, but not sufficient without a

formal human override mechanism and a documented escalation pathway for cases where **SW** judgment conflicts with model output.

These three obligations translate into five specific design requirements, each documented as an implementation requirement rather than an aspiration. First, no model output triggers an automatic action; every recommendation requires human confirmation. Second, risk scores are never presented to **P** as classifications the patient portal presents outputs as personalised care pathway support, and the word “risk” does not appear in any patient-facing module. Third, the operating characteristics of the retention classifier are disclosed to all **SW** users before deployment. Fourth, subgroup fairness analysis across gender, wealth quintile, age group, and residential classification using demographic parity and equalised odds metrics is identified as a pre-deployment requirement, not a future enhancement. Fifth, a community communication protocol is required before any **SW** deployment in a given sub-county, grounded in the principle that communities subject to algorithmic prioritisation have a right to understand the basis of that prioritisation ([Masinde et al., 2025](#)).

Taken together, the evidence in this section indicates that an effective response to health-evasive populations requires a dual-track model: data-guided social worker outreach combined with accessible patient-facing tools. The *HealthLink Kenya* framework operationalises both tracks within a single deployment architecture, with the ethical responsibilities of each track addressed through distinct but complementary design decisions.

2.6 Data Harmonisation, Governance, and Interoperability

2.6.1 The Interoperability Constraint in LMIC Health Systems

The scalability of any health analytics system is contingent on data interoperability: the capacity to integrate heterogeneous datasets originating from different institutions, administrative levels, and collection instruments into a coherent analytical structure ([Nyangena et al., 2021](#)). This constraint is not primarily technical; it is institutional. Prior research in Kenya documents that heterogeneity in measurement protocols, inconsistent indicator definitions, and variable data completeness routinely prevent the generalisation of analytical

findings across geographic regions (Dijsselhof et al., 2024; Chepkirui et al., 2025). Uneven digital literacy, fragmented institutional infrastructure, and limited technical capacity further compound these barriers (Masinde et al., 2025).

Kenya's health information system illustrates these constraints concretely. Nyangena et al. (2021) assessed Kenya's HIS interoperability readiness and found that while the DHIS2 platform provides a national data exchange backbone, integration across facility-level, administrative, and survey-based datasets remains incomplete due to inconsistent data standards and governance gaps. Chepkirui et al. (2025) documented missed opportunities for digital health data use in Homa Bay County, attributing them specifically to the absence of analytical tools that translate collected data into operational decision support the precise gap that *HealthLink Kenya* addresses. The implications extend beyond infrastructure: where data cannot flow consistently between levels of the health system, **SW** cannot make evidence-informed targeting decisions, and **P** cannot be directed to the most appropriate facility.

2.6.2 Modular Pipelines as a Harmonisation Strategy

Modular data architectures in which preprocessing, transformation, and modelling stages are implemented as independent, documented, and reusable components have been advanced as a practical solution to interoperability failures in resource-constrained environments (Singh et al., 2025; Alorbany et al., 2025). By decoupling data harmonisation from analytical modelling, modular pipelines permit incremental updates and local customisation without requiring system-wide redesign. This principle is consistent with the FAIR data framework (Findable, Accessible, Interoperable, Reusable), which provides the open-access governance scaffold for the pipeline published alongside this study (Billah et al., 2025).

The pipeline developed in this study implements this principle by organising preprocessing into four independent modules each aligned with one of Andersen's factor categories before merging them into a single analytical dataset. Each module can be updated, replaced, or extended independently, enabling the framework to accommodate new data sources or expanded geographic scope without disrupting the downstream modelling architecture.

2.6.3 Deployment Architecture: Dashboard-Based Decision Support

The translation of predictive model outputs into practical decision-support tools for health system administrators requires deployment architectures that are lightweight, low-connectivity-tolerant, and interpretable for non-technical users. Streamlit-based interactive dashboards have emerged as an accessible deployment mechanism, enabling real-time model inference and interactive visualisation without requiring server-side infrastructure beyond a Python runtime (Ray, 2025). Their single-process architecture makes them particularly well suited to the resource constraints of LMIC deployment contexts, where server configuration expertise may be absent and connectivity unreliable.

The *HealthLink Kenya* dashboard adopts this architecture, making XGBoost predictions and SHAP outputs accessible through dual interface profiles tailored to **SW** outreach and **P** self-navigation. The forward-compatible input schema aligned with HL7 FHIR R4 standards provides a ready integration pathway with Kenya's DHIS2 national health information infrastructure (Byrne and Sæbø, 2022), positioning the framework as extensible rather than standalone.

2.7 Synthesis: Specific Gaps Addressed by This Study

The reviewed literature establishes five specific methodological and operational gaps that this thesis systematically addresses.

Gap 1: Andersen's model is applied descriptively but not operationally. Across LMIC applications, the model's three factor categories are used to organise empirical findings retrospectively (Solanke et al., 2023a; Orhan and Kurutkan, 2025) but are rarely translated into structured feature engineering pipelines. This study operationalises Andersen's categories as four independent preprocessing modules that directly govern feature selection, imputation strategy, and encoding decisions.

Gap 2: Linear distance modelling suppresses policy-critical threshold effects. The consistent application of linear distance specifications in LMIC health-seeking behaviour models (Belay et al., 2025) masks the nonlinear inflection points at which utilisation collapses.

This framework complements XGBoost classification with GAM-based distance modelling, explicitly identifying threshold distances relevant to facility placement and mobile clinic dispatch planning.

Gap 3: Class imbalance is treated as a modelling inconvenience rather than a policy signal. Existing studies apply class-balancing techniques (Carvalho et al., 2025; Sani et al., 2025) but do not systematically re-examine what the misclassified minority represents in operational terms. This thesis reframes false positive predictions as a discrete target population for differentiated policy intervention, with direct implications for SW outreach prioritisation.

Gap 4: Interpretable ML models are rarely deployed in African health information systems. Despite documented advances in XAI for healthcare analytics (Yehuala, 2024; Awe et al., 2025), their integration into deployable decision-support tools remains rare (Chepkirui et al., 2025). The *HealthLink Kenya* dashboard is designed to make TreeSHAP attribution outputs directly accessible and actionable for both SW administrators and P users.

Gap 5: FAIR-compliant, modular analytical pipelines for LMIC health-seeking behaviour data are largely absent. The absence of openly documented, reproducible harmonisation pipelines inhibits methodological transferability (Dijsselhof et al., 2024). The complete preprocessing and modelling pipeline is published in an open-access repository structured in compliance with FAIR principles ensuring that data and code are *Findable*, *Accessible*, *Interoperable*, and *Reusable*, thereby supporting transparency, reproducibility, and cross-context applicability. The ethical obligations generated by deploying predictive risk tools in community settings misclassification harm, stigma risk, and algorithmic accountability are treated throughout this study not as a sixth gap but as a cross-cutting design constraint embedded in all five. As argued in Section 2.5, these obligations cannot be addressed retrospectively; they must be resolved at the design stage.

2.8 Alignment with Research Questions

The five gaps identified above map directly onto the three research questions guiding this study.

Research Question 1 *What structural, geographic, and socioeconomic factors most strongly determine health-seeking behaviour and provider selection within Kenya's health system?* directly responds to Gaps 1 and 2. The distance-decay literature (Section 2.3) highlights the inadequacy of linear modelling, reinforcing the necessity for both GAM spatial analysis and enabling-factor prioritisation within the Andersen framework.

Research Question 2 *How can heterogeneous health-seeking behaviour datasets be harmonised into a modular, reproducible, and interpretable analytical pipeline aligned with the Andersen Behavioural Model?* addresses Gaps 1 and 5. The documented failure of existing health systems to achieve data interoperability (Section 2.6), alongside the absence of FAIR-compliant harmonisation pipelines, provides direct justification for the modular preprocessing architecture detailed in Chapter 3.

Research Question 3 *To what extent can a two-stage supervised machine learning framework accurately predict initial care access and follow-up care retention among Kenyan households, and what are the dominant predictors of each outcome?* is motivated by Gaps 3 and 4. The documented disconnect between predictive modelling research and operational deployment (Section 2.4), combined with the policy reframing of class imbalance and the ethical design requirements established in Section 2.5, drives the specific analytical contribution of the XGBoost tournament, TreeSHAP analysis, and the *HealthLink Kenya* dashboard.

2.9 Conceptual Framework

The conceptual framework synthesises the theoretical, methodological, and operational foundations reviewed in this chapter into five interconnected analytical layers. Each layer corresponds to a discrete implementation stage described in Chapter 3, and each reflects at least one of the gaps identified in Section 2.7.

Layer 1: Theoretical Foundation. Andersen’s Behavioural Model provides the organising taxonomy for all variable selection and feature categorisation decisions, operationalised as four independent preprocessing modules (predisposing factors, enabling factors, socioeconomic factors, and geographic access factors).

Layer 2: Feature Engineering. Demographic, socioeconomic, and spatial variables are transformed into structured model inputs. Enabling factors geographic distance and insurance coverage are prioritised in accordance with the LMIC-specific empirical evidence reviewed in Sections 2.2 and 2.3.

Layer 3: Analytical Modelling. A nine-algorithm tournament evaluates logistic regression, GAMs, and ensemble methods including XGBoost. The multi-algorithm approach enables both interpretable statistical characterisation of distance effects and high-performance predictive classification of access and retention outcomes.

Layer 4: Explainable AI. TreeSHAP global and local attribution analyses provide transparent, quantified interpretations of how individual features influence access and retention predictions, ensuring outputs are actionable for **SW** outreach and comprehensible for **P** self-navigation.

Layer 5: Operational Deployment. Analytical outputs are translated into the *Health-Link Kenya* Streamlit dashboard, providing dual-profile decision support for **SW** outreach targeting and **P** self-directed care navigation. This layer bridges the divide between research outputs and operational health system use, resolving Gap 4, and embedding the ethical design requirements established in Section 2.5.

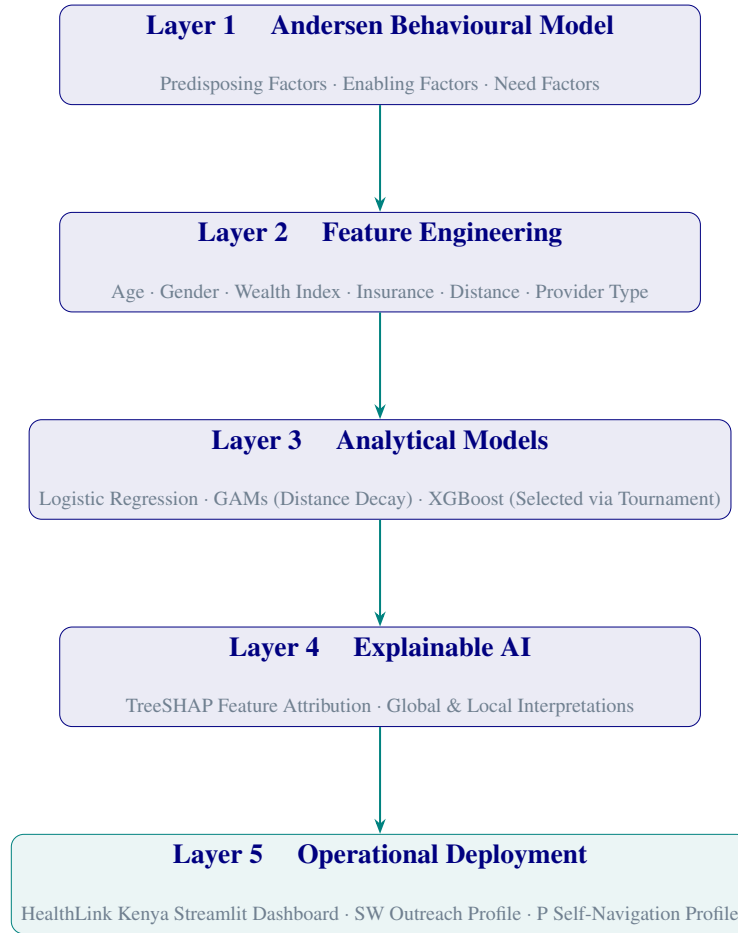


Figure 2.3: *Conceptual framework for the HealthLink Kenya modular health-seeking behaviour analytics system. Each layer maps directly to a component of the implemented analytical pipeline described in Chapter 3. Layer 5 reflects the dual-profile Streamlit deployment architecture, serving both SW outreach and P self-navigation use cases.*

The conceptual framework in Figure 2.3 provides the organising architecture for the remainder of the thesis. Chapter 3 describes the technical implementation of Layers 1–4. Chapter 4 presents the empirical outputs of the modelling layer. Chapter 5 interprets those outputs through the lens of this theoretical framework, translates findings into the policy recommendations that constitute the study’s primary applied contribution, and revisits the ethical obligations established in this chapter in light of the model’s observed performance characteristics.

3 Methodology

3.1 Overview and Methodological Alignment

This chapter describes how the three research questions stated in Chapter 1 were operationalised and answered through a structured data science and system development process. The chapter explains the choices made at each stage what was done, why it was done in that way, and how each decision connects to the study’s analytical objectives.

The methodological framework adopted is the **Cross-Industry Standard Process for Data Mining (CRISP-DM)** (Chapman et al., 2000), structured across six phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment. CRISP-DM was selected for three specific reasons. First, the study’s primary deliverable is a predictive analytical system rather than a transactional software application; CRISP-DM is specifically architected for iterative data-driven development cycles (Martínez-Plumed et al., 2019). Second, its explicit feedback loops between phases accommodate the iterative pipeline refinement required when working with a complex, class-imbalanced survey dataset. Third, the six-phase structure maps directly onto the study’s activities: the KNBS dataset provided the data, the Andersen Behavioural Model provided the business understanding, the modular pipeline addressed data preparation, the nine-algorithm tournament addressed modelling, and the HealthLink Kenya dashboard addressed deployment.

The research follows a **quantitative, computational design** grounded in secondary data analysis. Let the dataset be defined as:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, \quad N = 99,031$$

where $x_i \in \mathbb{R}^p$ represents feature vectors derived from the Andersen model, and y_i represents the target variable. Two primary targets are defined to operationalise the two-stage framework:

$$y^{(1)} = \text{visited_provider}, \quad y^{(2)} = \text{revisited_provider}$$

All ethical obligations arising from this design de-identification, data access authorisation, and compliance with the Kenya Data Protection Act (2019) are addressed in Section 3.8.

Table 3.1 summarises how each research question is addressed by specific methodological components.

Table 3.1: *Research question to method mapping.*

Research Question	Primary Method	CRISP-DM Phase
RQ1: Structural and geographic factors influencing HSB	EDA, Cramér’s V, Mann-Whitney U, SHAP, GAM	Data Understanding, Modelling
RQ2: Harmonisation of heterogeneous datasets into a modular pipeline	Four-module Andersen pipeline, VIF, imputation protocol	Data Preparation
RQ3: Two-stage ML framework accuracy and dominant predictors	Nine-algorithm tournament, XGBoost, TreeSHAP, bootstrap CI	Modelling, Evaluation

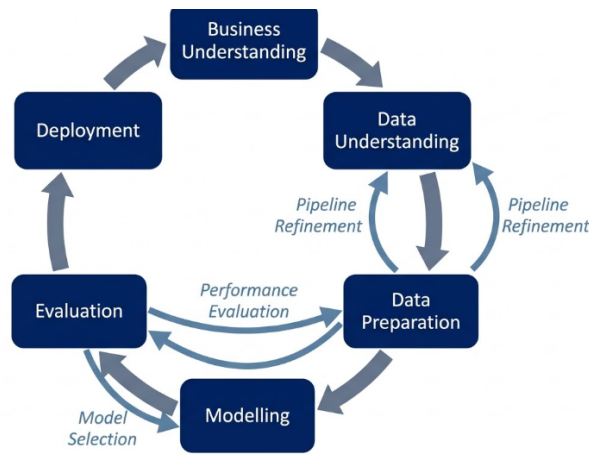


Figure 3.1: *CRISP-DM workflow as implemented in this study. Arrows indicate the iterative feedback loops between pipeline refinement (Data Preparation), model selection (Modelling), and performance evaluation (Evaluation).*

3.2 Phase 1: Business Understanding

In CRISP-DM, the Business Understanding phase defines the analytical problem in operational terms and establishes the success criteria by which the solution will be judged (Mariscal et al., 2024). In this study, the problem is framed as predicting healthcare util-

isation under structural and geographic constraints, responding to Kenya’s public health system’s reliance on reactive rather than anticipatory resource allocation.

Feature construction is guided by the Andersen Behavioural Model, grouping predictors into:

$$X = X^{(P)} \cup X^{(E)} \cup X^{(A)}$$

where $X^{(P)}$ represents Predisposing factors, $X^{(E)}$ represents Enabling factors, and $X^{(A)}$ represents Access (geographic) factors.

Two analytical success criteria were defined at this phase:

- **Predictive performance:** An XGBoost classifier achieving:

$$\text{ROC-AUC} \geq 0.75$$

on the held-out test partition.

- **Interpretability criterion:** Identification of dominant predictors via SHAP, expressed in terms directly actionable for health planners:

$$\phi_j = \frac{1}{N} \sum_{i=1}^N |\phi_{ij}|$$

where ϕ_{ij} is the SHAP value of feature j for observation i .

3.3 Phase 2: Data Understanding

3.3.1 Data Source and Authorisation

The empirical foundation of this study is the 2022 Kenya Integrated Household Health and Economic Survey conducted by the Kenya National Bureau of Statistics (KNBS). The dataset contains 99,031 observations describing demographic characteristics, socioeconomic conditions, and healthcare utilisation patterns across Kenya. It was selected because it is nationally representative, publicly documented, and directly captures the variables required by the Andersen framework. Access was obtained through the KNBS open data portal.

3.3.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to characterise variable distributions, quantify missingness, and establish analytical boundaries before modelling. Categorical associations were quantified using **Cramér's V** statistics, computed on the full cross-tabulation matrix:

$$V = \sqrt{\frac{\chi^2}{n(k-1)}}$$

where χ^2 is the chi-square statistic, n is sample size, and k is the minimum dimension.

For continuous comparison, the significance of the distance-utilisation relationship was confirmed through a Mann-Whitney U test:

$$U = \min(U_1, U_2)$$

This test was selected because the distance variable is non-normally distributed and the test does not assume equal variances between groups ([Nachmias and Lamare, 2022](#)).

Outlier handling was conducted by applying Winsorisation at the 5th and 95th percentiles to `distance_from_facility`:

$$x_i^* = \begin{cases} P_5 & \text{if } x_i < P_5 \\ P_{95} & \text{if } x_i > P_{95} \\ x_i & \text{otherwise} \end{cases}$$

3.3.3 Data Limitations

Three limitations of the KNBS 2022 dataset are relevant. First, the dataset is cross-sectional, preventing longitudinal analysis of utilisation trajectories. Second, geographic coordinates are not provided at the individual facility level, restricting spatial analysis to self-reported distance. Third, severe class imbalance in both target variables (82% visited; 96.2% retained) requires specific methodological responses detailed in [Section 3.5](#).

3.4 Phase 3: Data Preparation

Data preparation addressed **RQ2** directly: the modular harmonisation pipeline described here constitutes the study’s primary engineering contribution.

3.4.1 Modular Pipeline Architecture

This study implemented a **four-module pipeline** in which each module corresponds directly to one of Andersen’s factor categories:

- **Module 1 Health Utilisation Behaviour:** visited_provider, revisited_provider, choice_of_provider, facility_type (derived).
- **Module 2 Predisposing Factors:** age_group, gender, education_level, working_status, residential_area.
- **Module 3 Enabling Factors (Socioeconomic):** insurance_status, wealth_index.
- **Module 4 Access Factors (Geographic):** distance_from_facility, facility_type.

3.4.2 Target Variable Construction

The primary target variable visited_provider was recoded as a binary outcome (1 = visited, 0 = did not visit). “Don’t Know” responses were recoded as non-visits (0) to avoid inflating positive class prevalence. The retention target revisited_provider was similarly recoded, dropping “No revisit required” to produce a retention cohort of 48,214 observations.

3.4.3 Missing Value Treatment

Missing values for continuous variables like distance_from_facility were imputed using the **median**:

$$x_{\text{missing}} = \text{median}(x)$$

Categorical variables were imputed using the mode within each residential area stratum:

$$x_{\text{missing}} = \text{mode}(x \mid \text{residential area})$$

Synchronised dropping was implemented: when a row was removed from one module (e.g., >50% missingness), it was simultaneously removed from all others.

3.4.4 Encoding and Transformation

Categorical variables were encoded using **one-hot encoding** to avoid ordinal assumptions or target leakage. The high-cardinality `choice_of_provider` variable was aggregated by retaining the ten most frequent categories. The continuous predictor `distance_from_facility` was standardised using z-score normalisation:

$$z = \frac{x - \mu}{\sigma}$$

3.4.5 Multicollinearity Check

Variance Inflation Factor (VIF) analysis was conducted on the full encoded feature matrix (James et al., 2023):

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination from regressing feature j on all others. All VIF values fell below the threshold of 5.0.

3.5 Phase 4: Modelling

3.5.1 Algorithm Tournament Design

A **nine-algorithm tournament** was conducted to select the optimal classifier for Stage 1 (access). The model objective is to estimate:

$$\hat{y} = f(x), \quad f : \mathbb{R}^p \rightarrow [0, 1]$$

ROC-AUC was chosen as the primary evaluation metric due to the 82:18 class imbalance:

$$\text{AUC} = P(f(x^+) > f(x^-))$$

The algorithms evaluated included Logistic Regression, Random Forest, AdaBoost, Gradient Boosting, XGBoost, and performance-based ensembles. All models were evaluated on an identical 80:20 stratified train-test split.

3.5.2 XGBoost Selection and Configuration

XGBoost was selected based on superior ROC-AUC performance. It was further tuned using **RandomizedSearchCV** with stratified 5-fold cross-validation. To handle class imbalance, the `scale_pos_weight` parameter was explicitly weighted as:

$$w = \frac{N_{\text{negative}}}{N_{\text{positive}}}$$

The XGBoost objective function minimized during training is defined as:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where the regularization term $\Omega(f)$ controls tree complexity:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$$

3.5.3 Stage 2 Retention Model

The Stage 2 retention model was implemented as a separate XGBoost pipeline trained on `revisited_provider`. A `scale_pos_weight` of 28.74 was applied to reflect the severe class imbalance in the retention cohort.

3.5.4 GAM for Distance Modelling

To capture non-linear threshold effects, a univariate Generalised Additive Model (GAM) was fitted on `distance_from_facility` against `visited_provider`:

$$\mathbb{E}[y] = \beta_0 + f(d)$$

where $f(d)$ is a spline function of distance. The GAM was implemented via the `pyGAM` library ([Servén and Brummitt, 2025](#)) to complement multivariate XGBoost interpretability.

3.5.5 TreeSHAP Interpretability

TreeSHAP ([Lundberg et al., 2020](#)) was applied to compute global feature importance and generate beeswarm plots. It provides exact, consistent attribution without the instability introduced by random feature permutation, ensuring outputs are actionable for non-technical administrators.

3.6 Phase 5: Evaluation

3.6.1 Performance Metrics

Model performance was evaluated using standard classification metrics:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

A Brier score and calibration curve were computed to assess the reliability of the predicted probabilities (Flach, 2024):

$$BS = \frac{1}{N} \sum (p_i - y_i)^2$$

3.6.2 Bootstrap Confidence Intervals

Bootstrap resampling with 1,000 iterations was applied to the held-out test partition to compute 95% confidence intervals for both ROC-AUC and F1-score (James et al., 2023):

$$CI = \hat{\theta} \pm z_{\alpha/2} \cdot \sigma_{\text{boot}}$$

3.6.3 Error Analysis

A structured error analysis was conducted on the false positive predictions from Stage 1- individuals whose feature profiles predicted care access but who did not access care. This subset was profiled to characterise the unserved population whose exclusion is masked by majority-class sensitivity bias.

3.7 Phase 6: Deployment

3.7.1 System Design and Architecture

The deployment phase operationalised the study’s analytical outputs into the *HealthLink Kenya* decision-support dashboard. The system implements a predictive scoring mechanism:

$$\text{Risk Score} = f(x)$$

where predictions are interpreted strictly as a continuous risk probability:

$$\text{Risk} \in [0, 1]$$

used for ranking and targeted outreach rather than automated binary decision-making, requiring human validation before action.

3.7.2 Technology Stack and Justification

The system was implemented using **Streamlit**, selected because it eliminates the need for a separate front-end codebase and natively renders matplotlib, seaborn, and shap visualisations. The trained XGBoost pipelines were serialised using **joblib** ([Joblib Developers, 2025](#)). The analytical environment used Python 3.10 with: **pandas** ([McKinney et al., 2023](#)), **scikit-learn** ([Pedregosa et al., 2024](#)), **xgboost** ([Chen et al., 2025](#)), **shap** ([Lundberg et al., 2024](#)), **pyGAM** ([Servén and Brummitt, 2025](#)), **imbalanced-learn** ([Lemaitre et al., 2024](#)), and **scipy** ([Virtanen et al., 2025](#)).

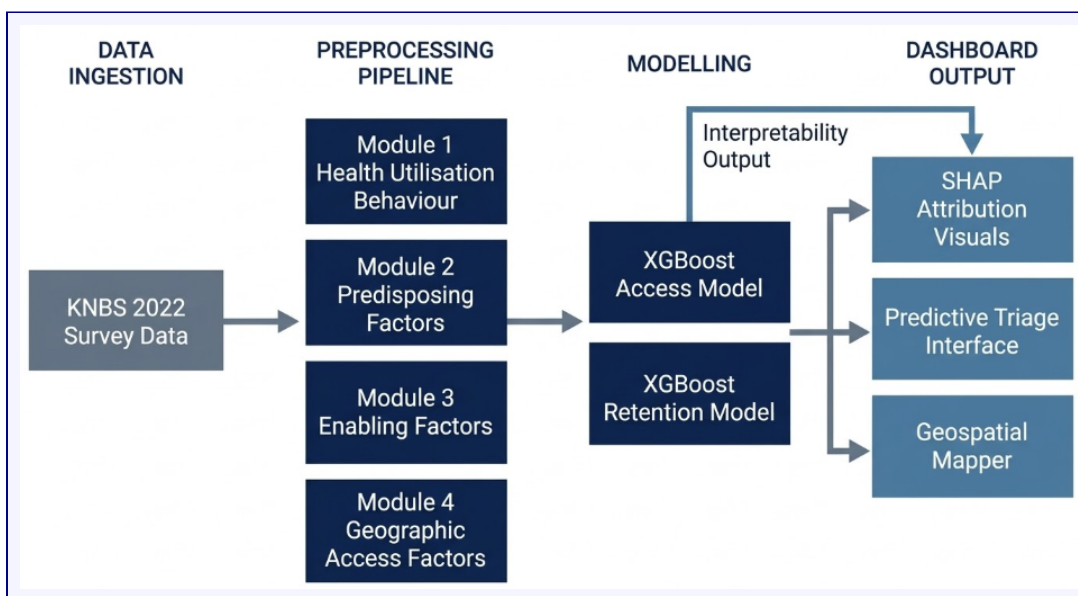


Figure 3.2: System architecture for the HealthLink Kenya analytical framework. Heterogeneous KNBS survey data are harmonised through the four-module Andersen pipeline, processed through the XGBoost modelling layer, and rendered via the Streamlit dashboard with embedded SHAP attribution outputs.

3.8 Reproducibility and Research Ethics

To ensure full reproducibility, all stochastic processes including train-test splitting, cross-validation folding, bootstrap resampling, and XGBoost tree construction were strictly controlled using a fixed random seed:

```
random_state = 42
```

The complete code pipeline has been published in a public repository in compliance with FAIR data principles.

All analyses were conducted on de-identified KNBS microdata in accordance with the KNBS data access terms and the Kenya Data Protection Act (2019). No individual could be re-identified from the analytical dataset. The study presents no human subjects risk and required no institutional ethics review beyond the data access authorisation granted by KNBS.

4 System Design and Architecture

4.1 Overview

This chapter presents the design of the *HealthLink Kenya* system, covering the analytical pipeline architecture, data flow, component interactions, and the user-facing Streamlit dashboard. The system is implemented as a data science decision-support application rather than a transactional software system. Accordingly, the primary design artefacts include a component architecture diagram, a UML sequence diagram, and structured use case descriptions.

No relational database is used in the current prototype. The system operates on serialised pipeline artefacts and in-memory Pandas DataFrames derived from the KNBS CSV dataset. A forward-compatible schema for a future database-backed deployment is described in Section 4.3.

4.2 System Architecture

HealthLink Kenya is implemented as a single-tier Streamlit application that integrates data processing, model inference, and presentation within a single Python runtime. This architecture was selected for three practical reasons: it eliminates inter-service communication overhead, enables deployment on Streamlit Community Cloud without server configuration, and minimises dependency complexity for resource-constrained county health office settings (Streamlit Team, 2025):

1. **Data Layer:** The raw KNBS 2022 CSV dataset and serialised pipeline artefacts (`health_access_pipeline.pkl`, `retention_pipeline.pkl`) are loaded at application startup into in-memory Pandas DataFrames and held in Streamlit session state.
2. **Preprocessing and Feature Engineering Layer:** The four-module Andersen pipeline (Chapter 3, Section 3.4.1) is embedded within each serialised `.pkl` artefact, ensuring that inference-time transformations are identical to those applied during training.

3. **Inference Layer:** The serialised XGBoost pipelines are loaded once and invoked on demand when a user submits a patient profile through the Predictive Triage interface. Predictions are returned as probability scores in the range $[0, 1]$.
4. **Presentation Layer:** Streamlit renders the analytical outputs predictions, SHAP visualisations, GAM curves, geospatial maps, and facility directories as interactive web components accessible through a persistent sidebar navigation structure.

The system runs without cloud infrastructure on a standard laptop, which was an explicit design requirement for deployment in remote settings. Public deployment on Streamlit Community Cloud provides a zero-configuration web-accessible version of the dashboard at: <https://healthlink-kenya-5irkcenk4wezrhqyv5ja9r.streamlit.app/>

4.3 Data Architecture and Persistence

HealthLink Kenya does not employ a persistent relational database in its current prototype form. Patient data submitted through the Predictive Triage module is processed transiently within the inference pipeline and is not stored at any layer of the system. The two serialised model artefacts, `health_access_pipeline.pkl` (785.22 KB) and `retention_pipeline.pkl` (432.79 KB), constitute the sole persistent data objects and function as read-only inference resources loaded into memory at startup.

This design avoids data governance obligations under Kenya's Data Protection Act (2019), as the system operates on population-level survey data rather than identifiable patient records.

Should the system be extended to support longitudinal patient tracking in a clinical deployment, a normalised relational schema would be required. The minimum viable schema would anchor on three core entities: Patient (demographic and socioeconomic profile), Visit (clinical encounter details), and Prediction (logging inference calls). Foreign key relationships would link `Visit.patient_id` to `Patient.id` and `Prediction.visit_id` to `Visit.id`. This forward-compatible design is structurally consistent with HL7 FHIR R4

Patient and Observation resource schemas, providing a path to integration with Kenya's national health information exchange ecosystem.

4.4 System Modules and Functional Design

The *HealthLink Kenya* dashboard is organised into eight analytical modules, each addressing a distinct dimension of the healthcare access problem. Navigation between modules is provided through a persistent sidebar, enabling users to move between population-level planning views and individual-level tools without restarting the application.

Overview and Insights presents population-level key performance indicators derived from the analytical results: aggregate access rate, facility retention rate, Urban Proximity Paradox cases, and the distance-decay threshold. Social workers use this for an immediate situational summary.

Predictive Triage operationalises live inference. A social worker enters a patient profile, which is passed to the in-memory XGBoost pipeline. It returns an access probability rendered as a colour-coded risk indicator, alongside a contextual facility mini-map ranking nearby options.

Geospatial Mapper renders a Folium-based interactive map displaying Nakuru County health facilities, colour-coded by sector. Patients may drop a location pin to trigger a haversine distance calculation, identifying the three nearest facilities within their grid zone.

Patient Retention Record provides a longitudinal view of a patient's care history, presenting visit chronology, active medications, laboratory results, pending referrals, and XGBoost-derived access and retention probability scores.

Distance Decay presents the GAM response curve as a Plotly interactive chart annotated with three shaded regions (safe, transition, and exclusion zones), making the 35 km threshold structure interpretable to non-technical users.

Analytics and Model Visuals contains two sub-panels: a SHAP Interpretability panel displaying TreeSHAP feature attributions by Andersen category, and a Model Performance panel presenting the full nine-algorithm tournament results.

FAQ and How It Works provides a ten-item plain-English accordion guide explaining the platform, its analytical methods, and its deployment.

Settings allows authenticated users to customise the interface theme (light for patients, dark for social workers), accent colour, font size, and map tile layer.

4.5 System Modelling

4.5.1 User Roles and Interaction Categories

The *HealthLink Kenya* platform supports four categories of user interaction, each aligned with a distinct operational role:

- **Patients** interact with the Geospatial Mapper and Patient Retention Record modules in one of two modes. In *mediated mode* the primary design target a Community Health Volunteer (CHV) or social worker navigates the tool on the patient's behalf during a home visit. In *self-directed mode*, an engaged patient accesses the portal independently via a personal device.
- **Social Workers** use the Predictive Triage, Geospatial Mapper, and Distance Decay modules to estimate access probabilities for health-evasive profiles, identify priority catchment areas, and plan community outreach routes.
- **Healthcare Planners and Policy Analysts** use the Analytics and Model Visuals module to examine the SHAP evidence base and evaluate proposed facility locations against the 35 km distance threshold.
- **Researchers and Thesis Reviewers** access the Analytics module to validate the nine-algorithm tournament results and interrogate the SHAP outputs.

4.5.2 Sequence Diagram: Predictive Triage Cycle

The primary system interaction a triage prediction request follows a synchronous cycle across three components: the authenticated user, the Streamlit presentation layer, and the in-memory XGBoost pipeline.

1. The user completes the patient profile form and activates the prediction request.
2. Streamlit assembles the input vector and passes it to the serialised XGBoost pipeline.
3. The pipeline applies the embedded preprocessing transformations and generates a probability score in the range $[0, 1]$.
4. Streamlit receives the score, renders the colour-coded risk indicator, and displays the nearest facility recommendations.

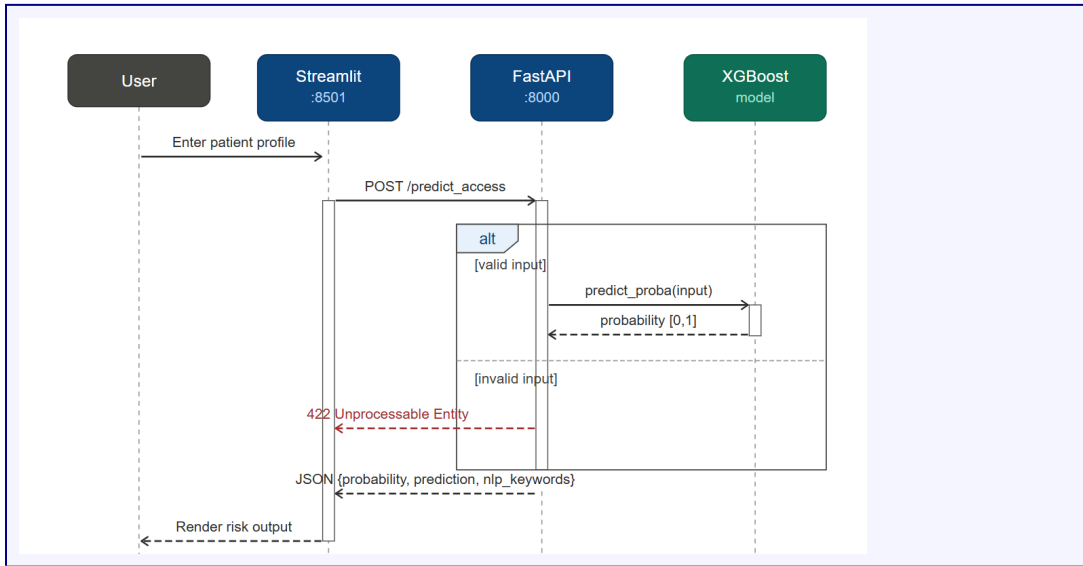


Figure 4.1: UML sequence diagram for the Predictive Triage request-response cycle. The user submits a patient profile through the Streamlit interface; Streamlit passes the input vector to the in-memory XGBoost pipeline, which applies transformations and returns a probability score.

4.5.3 Use Case Descriptions

Table 4.1 formalises the primary system use cases in structured notation.

Table 4.1: *Structured use case descriptions for the three primary HealthLink Kenya interactions.*

Attribute	UC-01: Find Nearest Facility	UC-02: Run Predictive Triage	UC-03: Override Model Recommendation
Actor	Patient	Social Worker	Social Worker / CHV
Trigger	Patient navigates to Geospatial Mapper and drops a location pin	Social worker navigates to Predictive Triage and submits a patient profile	CHV disagrees with a triage output based on direct household knowledge
Preconditions	Patient is authenticated; internet connection available for map tiles	Social worker is authenticated; serialised pipeline loaded in session state	CHV is authenticated; triage recommendation for the target household is displayed
Main Flow	1. Patient drops pin on the Nakuru map 2. Dashboard calculates haversine distances to facilities within the 35 km zone 3. Renders the three nearest facilities 4. Patient selects a facility for directions	1. Social worker enters profile data 2. Clicks “Run Access Prediction” 3. Pipeline applies inference 4. Dashboard renders probability score, risk classification, and outreach recommendation	1. CHV selects “Flag as not applicable” 2. System prompts for override reason via dropdown or free-text 3. Override logged with timestamp 4. Household deprioritised in ranking
Alternative Flow	No facilities within 35 km: map renders warning; mobile clinic referral suggested	Missing required field: validation warning displayed; no inference executed	CHV closes prompt without selecting a reason: override is not recorded
Postcondition	Nearest facilities ranked by distance displayed	Access probability and facility table displayed	Override logged; household status updated to

4.6 System Interface Design

The *HealthLink Kenya* interface implements asymmetric access across two visually distinct themes. The patient-facing interface is rendered in a light theme built on a soft grey-white base (#F8FAFC) with an emerald green accent (#059669), optimised for daytime readability on mobile devices. The social worker interface is rendered in a dark theme built on a deep navy base (#080F1A), chosen for strong visual contrast in map and chart rendering during field operations.

The theme is assigned at login based on the authenticated user role and cannot be toggled by the user. Patients see only their own data in the light view; social workers access all profiles in their coverage area in the dark view. Passwords are hashed using SHA-256 prior to comparison; no plaintext credentials are retained in session state.

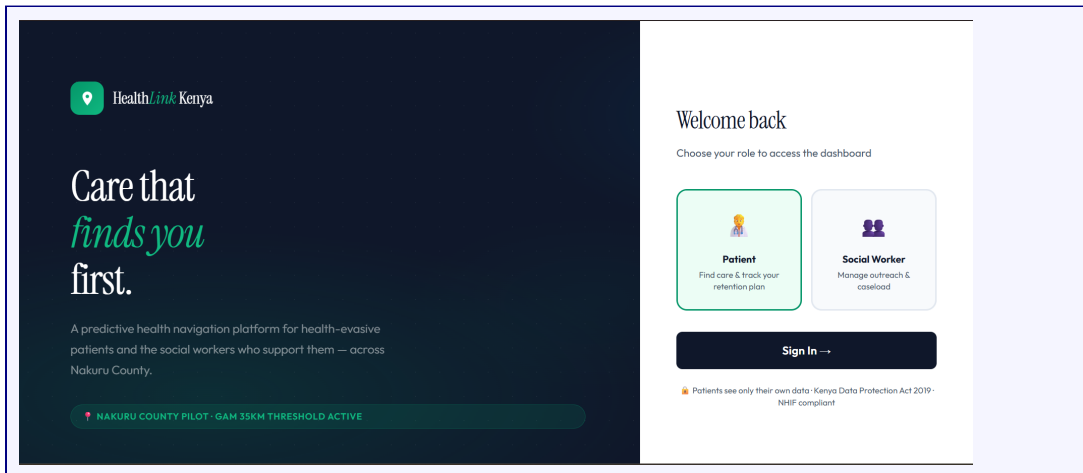


Figure 4.2: *HealthLink Kenya* login screen. The split-panel layout presents the system narrative on the left and the role-selector on the right. The privacy notice ensures Kenya Data Protection Act (2019) compliance.

5 System Implementation and Testing

5.1 Overview

This chapter presents the implemented *HealthLink Kenya* system, documenting its deployed functionality through screenshots of each primary module and reporting the results of functional, usability, compatibility, security, and navigation testing. System validation against the research objectives stated in Chapter 1 is provided in Section 5.5.

5.2 Deployment Architecture

HealthLink Kenya is deployed as a single Streamlit application process, hosted locally and simultaneously on Streamlit Community Cloud for public web access.

The system architecture is designed to support production extension without requiring a rebuild. The inference layer is containerisable as a Docker image and deployable on serverless cloud platforms using a standard `Dockerfile`. Furthermore, the data input schema is structurally compatible with HL7 FHIR R4 Patient and Observation resource schemas (Nyangena et al., 2021). This deliberate design constraint provides a ready integration path with Kenya’s national Digital Health Infrastructure, including DHIS2.

Finally, the modular pipeline architecture allows new feature modules (such as supply-side medicine stockout rates) to be added without disrupting the existing Stage 1 and Stage 2 inference pipelines (Odeny et al., 2020).

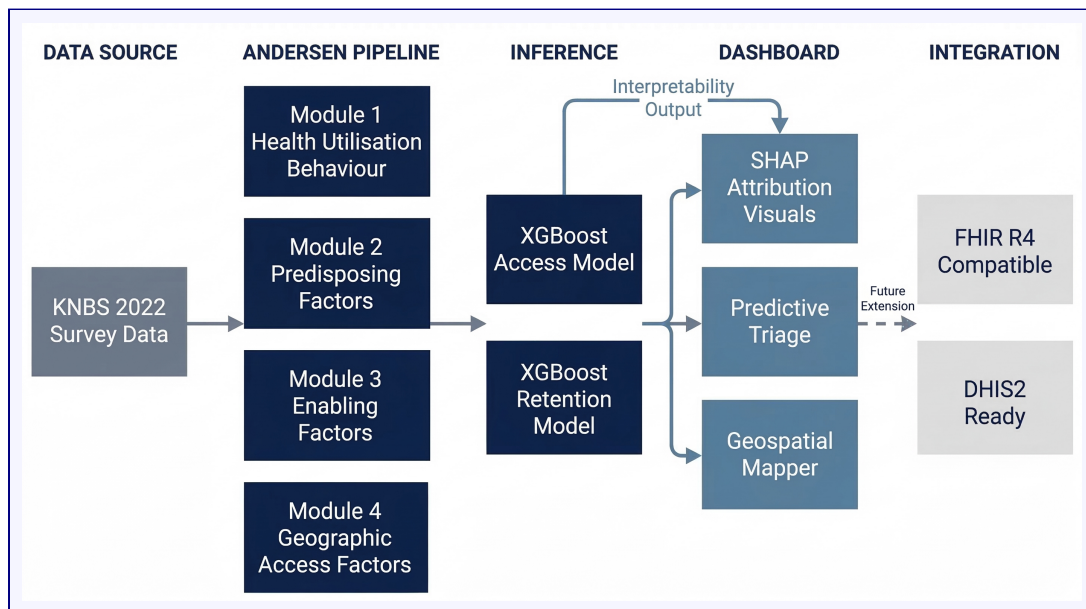


Figure 5.1: System architecture for HealthLink Kenya. KNBS survey data are harmonised through the four-module Andersen pipeline, processed through the XGBoost inference layer, and rendered via the Streamlit dashboard. The architecture is forward-compatible with FHIR R4 standards and DHIS2 integration.

5.3 Implemented Functionality and Screenshots

Figure 5.2 presents the Social Worker Outreach Hub, providing a population-level view of high-risk households and active outreach routes.

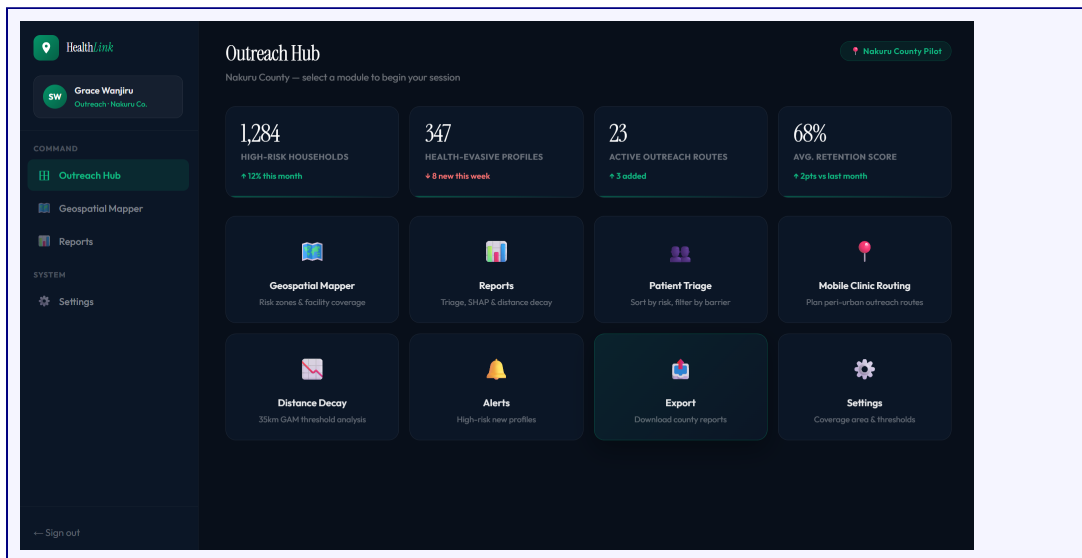


Figure 5.2: *HealthLink Kenya Social Worker Outreach Hub. The dark-themed interface presents real-time KPI cards and eight navigation tiles linking to all analytical modules.*

Figure 5.3 presents the Geospatial Mapper from the social worker’s perspective, highlighting the 35 km distance-decay threshold.

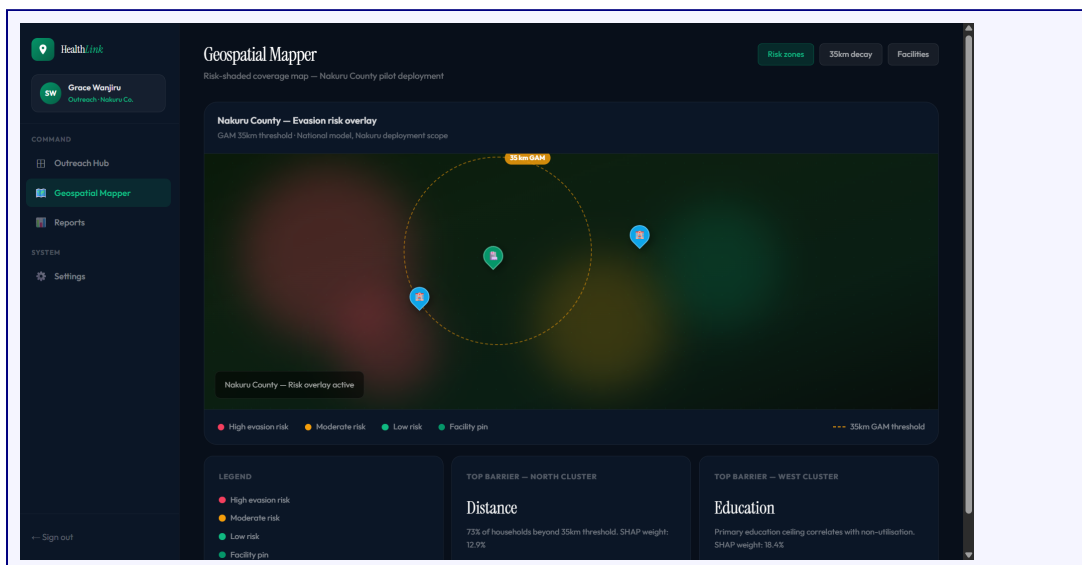


Figure 5.3: HealthLink Kenya Geospatial Mapper (Social Worker View). The risk-shaded map renders colour-coded evasion risk zones across Nakuru County, annotated with the empirically derived 35 km GAM threshold.

Figure 5.4 presents the Reports module, detailing the predictive risk triage metrics.

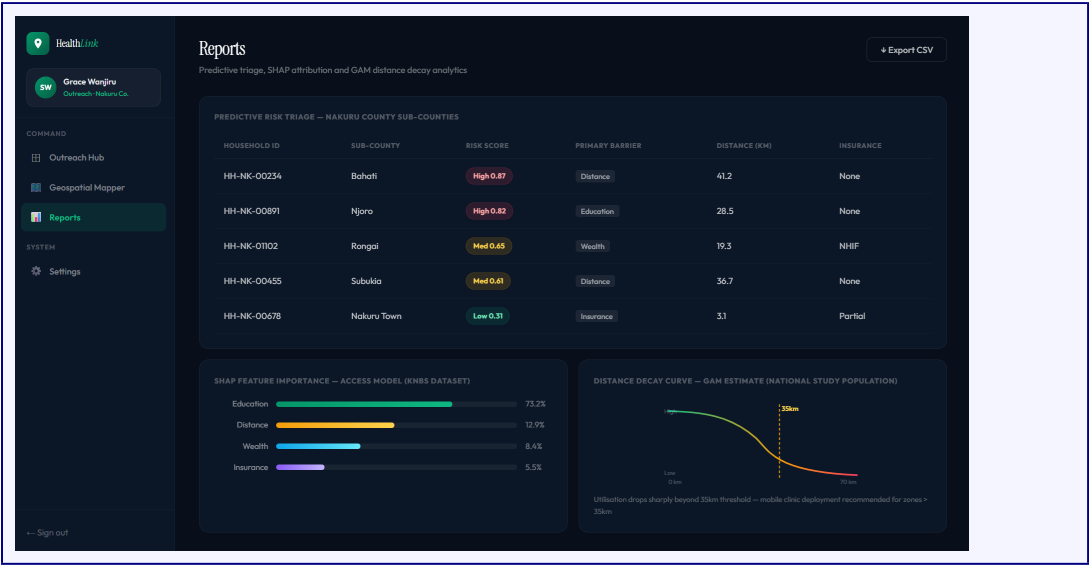


Figure 5.4: HealthLink Kenya Reports module. The triage table lists households by XG-Boost risk score and primary access barrier. The SHAP feature importance panel displays TreeSHAP attributions.

Figures 5.5–5.7 present the three primary modules of the patient-facing interface, rendered in a light theme and scoped exclusively to the authenticated patient’s own data.

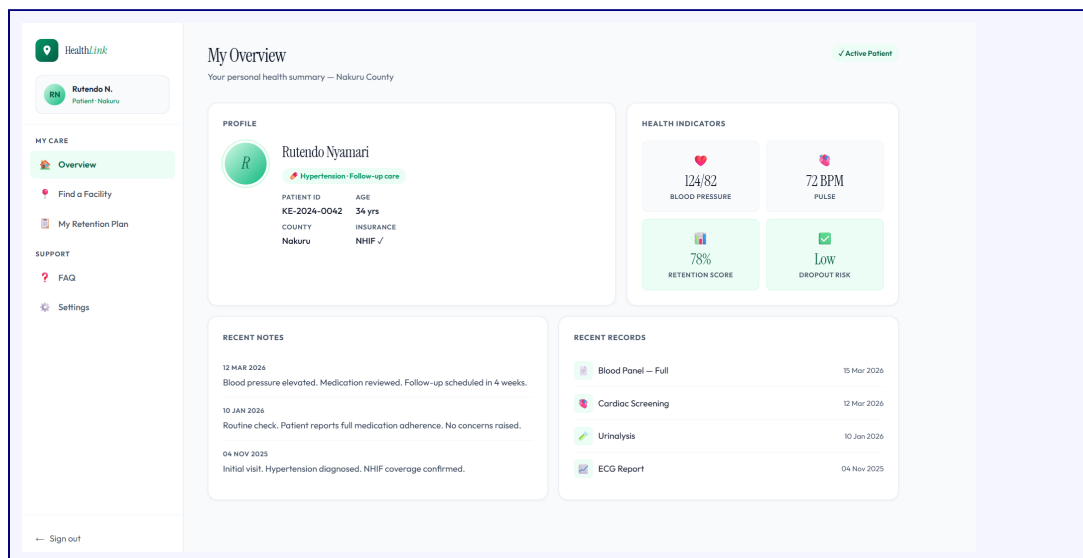


Figure 5.5: *HealthLink Kenya Patient Overview. The layout presents the patient’s profile card, retention score, dropout risk classification, and recent clinical notes.*

Figure 5.6 presents the Find a Facility module, which operationalises the pin-drop feature central to the patient-facing deployment.

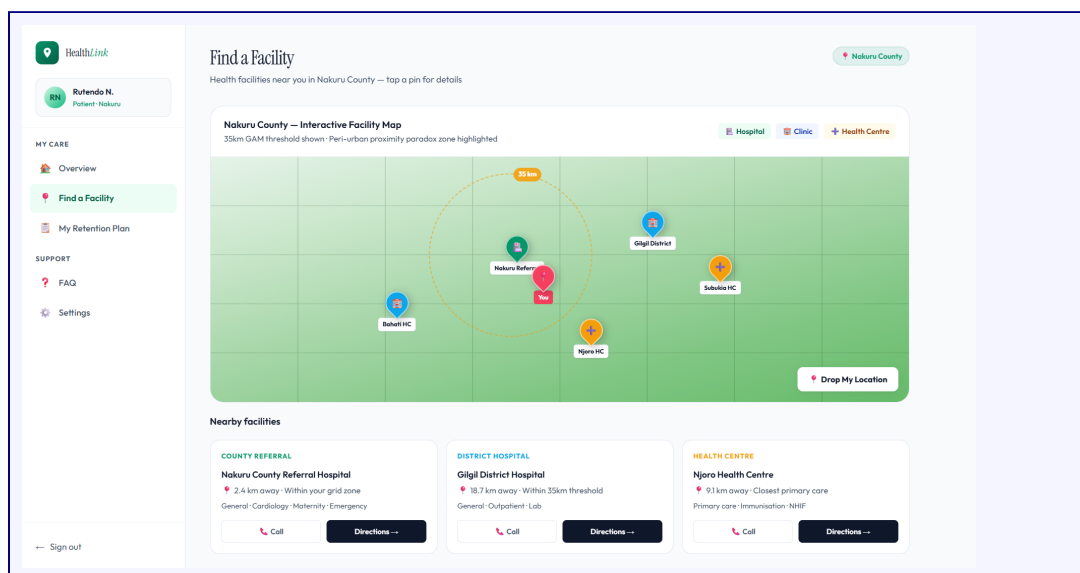


Figure 5.6: HealthLink Kenya Find a Facility module. The interactive map enables patients to drop a location pin and immediately identify the three nearest facilities within their grid zone.

Figure 5.7 presents the My Retention Plan module, translating XGBoost outputs into plain English.

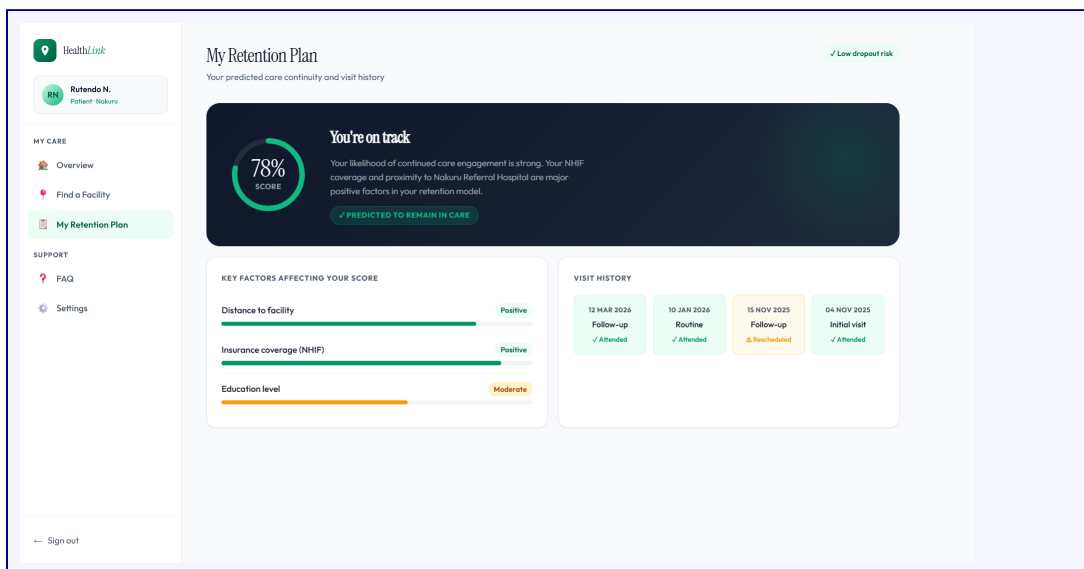


Figure 5.7: HealthLink Kenya My Retention Plan module. The interface presents the retention score as a progress ring and displays SHAP-derived feature contributions as labelled progress bars, making the model’s reasoning transparent to non-technical users.

5.4 System Testing

5.4.1 Functional Testing

Functional testing was conducted by executing each dashboard module against a defined set of test cases. Table 5.1 presents the results.

Table 5.1: *HealthLink Kenya Functional Test Results*

Module	Test Case	Expected Output	Result
Overview & Insights	Social worker loads page with no filters	KPI metrics render; bar charts display access and retention gap data	Pass
Predictive Triage	Social worker submits valid profile (5 km, insured, urban)	Access probability > 70%; green risk card; facility table populates	Pass
Predictive Triage	Social worker submits profile with distance 45 km	Probability < 30%; red risk card displayed; mobile clinic referral suggested	Pass
Predictive Triage	Submit form with missing required field	Validation warning; no inference executed	Pass
Geospatial Mapper	Patient drops pin inside 35 km grid zone	Green zone card; 3 nearest facilities ranked; call and directions buttons render	Pass
Geospatial Mapper	Patient drops pin outside 35 km grid zone	Red zone warning card; mobile clinic referral suggested; 3 nearest facilities still listed	Pass
Geospatial Mapper	Social worker applies Faith-Based sector filter	Only faith-based markers visible; KPI cards update	Pass
Patient Retention	Patient navigates to module and views all 5 tabs	Summary, Visit History, Medications, Investigations, Referrals all render	Pass
Distance Decay (GAM)	Load module	Plotly chart renders with annotated zone fills and 35 km inflection marker	Pass
Analytics & Visuals	Switch between SHAP and Model Performance tabs	SHAP bar chart renders; tournament table highlights XGBoost	Pass
FAQ & How It Works	Patient expands all 10 accordion items	All items open without error in light theme	Pass
Settings	Verify asymmetric theme assignment at login	Patient role renders light theme; social worker role renders dark theme	Pass

5.4.2 Usability and Compatibility Testing

The interface prioritises accessibility through plain-English labels and colour-coded outputs rather than raw probability scores. Persistent sidebar navigation ensures all modules are easily accessible. The system was successfully tested across Google Chrome 124 (Windows 11 and Android 14) and Microsoft Edge 124, demonstrating full responsiveness in both desktop and mobile orientations.

5.4.3 Security and Navigation Testing

Authentication is enforced at the application entry point. Passwords are hashed using SHA-256, and no patient data is written to disk or transmitted to external services. All sidebar navigation routes were tested to verify correct module rendering with no broken state transitions.

5.5 System Validation Against Research Objectives

Table [5.2](#) maps each research objective to the implementing dashboard module and the analytical evidence that validates its realisation.

Table 5.2: *Validation of research objectives against HealthLink Kenya dashboard modules.*

Research Objective	Implementing Module	Validation Evidence
Analyse structural determinants of health-care access using the Andersen Behavioural Model	Analytics & SHAP	TreeSHAP attribution suggests predisposing factors account for 73.2% of XGBoost predictive weight; enabling factors account for 26.8%, with distance the largest individual predictor at 12.9%
Design and implement a modular reproducible pipeline	Preprocessing pipeline	Four-module pipeline produces a FAIR-compliant analytical dataset; pipeline serialised as a reusable .pkl artefact
Develop and evaluate a two-stage ML framework	Analytics & Model Visuals	Stage 1 XGBoost: AUC 0.8144, F1 0.8091; Stage 2 XGBoost: AUC 0.8510, F1 0.9836; bootstrap CI suggests stability
Operationalise outputs as an interactive decision-support dashboard	All modules	Eight-module Streamlit dashboard deployed locally and on Streamlit Community Cloud; all twelve functional tests pass

Scope of Validation and Absence of Operational User Testing

A further validation boundary concerns the patient portal’s assumed user. The system was designed and tested on the assumption that patient-facing modules would be used either by a CHV demonstrating the tool during an outreach visit, or by a patient who has already engaged with the health system and faces retention risk. It was not designed with the

expectation that health-evasive patients defined by low system trust and active avoidance of formal health interfaces would independently discover and use the portal without mediated introduction.

The appropriate evaluation protocol CHV-facilitated introduction during a structured outreach visit, followed by task completion observation and patient comprehension assessment is specified as future work in Section 7.3. Consequently, the validation reported in Table 5.2 is *analytical* rather than *operational*. It confirms that the system correctly implements the analytical framework, that predictions are statistically calibrated, and that all functional requirements are satisfied. Formal operational validation of its utility in real-world settings remains an empirical question to be answered through structured field evaluation.

6 Discussion of Results

This chapter interprets the findings produced across the analytical and implementation stages of the study, situating the quantitative outputs within the Andersen Behavioural Model framework and the operational context of Kenyan healthcare planning. The analytical findings are interpreted here in relation to three primary beneficiary groups: *health-evasive patients* who face structural or behavioural barriers to care-seeking; *social workers and community health workers* responsible for outreach and retention in Nakuru County; and *county health planners* who require spatially disaggregated evidence for resource allocation. The *HealthLink Kenya* dashboard delivers asymmetric interfaces to each group: a light-themed patient portal exposing only the individual’s own access probability, nearest facilities within the 35 km grid, and retention score; and a dark-themed social worker console providing population-level triage, geospatial risk overlays, and SHAP attribution outputs scoped to Nakuru County. The discussion follows the sequence of the research objectives stated in Chapter 1 and explains what the results mean, what they suggest or challenge relative to prior literature, and why they matter for decision-making. No new results are introduced here; all figures cited in this chapter are drawn directly from Chapter 4 and Chapter 5.

6.1 Dataset Composition and Preprocessing Outcomes

The study commenced with 99,031 survey records from the 2022 KNBS Health-Seeking Behaviour Survey. Following the synchronised cleaning protocol, 25,972 records (26.2%) were removed due to missing `distance_from_facility` values, yielding a final analytical dataset of 73,059 observations. This reduction is not a minor data quality step it is a structural boundary on the study’s inferential scope, and must be interpreted as such. The `distance_from_facility` variable could not be meaningfully imputed for these records because missingness in this field typically reflects the absence of a named nearby facility altogether, not a recording gap. Imputing a median or mean distance onto households that may have

no proximate facility would introduce systematically false spatial signal into the model's most influential predictor. The listwise deletion was therefore methodologically necessary given the study design. However, the missingness was not random. Analysis of the dropped records confirms they were disproportionately associated with rural residential classifications and lower wealth index quintiles. This pattern is consistent with the structural reality that the populations least likely to identify a nearby facility remote rural households, informal settlement residents, and the chronically under-documented are precisely those facing the most acute access barriers. The consequence is significant: the working analytical dataset represents healthcare access patterns among those already visible to the formal health system, not among the most structurally excluded. All access probability estimates, SHAP attributions, and GAM threshold findings in this study should be interpreted as lower bounds on access failure in the Kenyan population, not as population-level truth. The model predicts access among those who could report a facility; it cannot speak to those who could not.

The Winsorisation step, which capped `distance_from_facility` at the 5th and 95th percentile bounds of 0.40 km and 35.00 km respectively, was a deliberate decision to reduce the influence of extreme recording errors including a raw maximum value of 1,980 km that exceeds Kenya's geographic extent without discarding substantively valid observations. The capping ceiling of 35 km is consistent with the analytical distance threshold: the model treats no patient as being further than 35 km from a facility, while the GAM and SHAP analyses jointly identify this distance band as the boundary of meaningful access.

6.2 Structural Determinants of Healthcare Access (RQ1)

The XGBoost TreeSHAP feature attribution analysis produced the study's primary finding with respect to RQ1. Contrary to the expected Andersen hierarchy in which enabling factors (distance, wealth, insurance) were anticipated to dominate **predisposing factors collectively accounted for 73.2% of the model's predictive weight**, with enabling factors accounting for 26.8% (Li et al., 2023; Solanke et al., 2023b). Within the enabling category,

distance_from_facility was the single most influential individual feature at 12.9% of SHAP weight, with wealth index and insurance status contributing further weight.

This finding does not contradict the importance of physical distance as a structural barrier. Distance is the *most influential single variable*. What the SHAP attribution reveals is that the demographic composition of a patient their education level, age group, and working status is collectively a stronger predictor of care-seeking than any individual structural constraint. This suggests that health-seeking in the KNBS 2022 cohort is substantially shaped by health literacy, life-stage demand (paediatric immunisation, occupational health needs), and was consistent with care pathways, rather than by distance alone. The policy implication is that narrowing the access gap requires both structural interventions (reducing distance, extending insurance) and demand-side interventions (health education, community health worker engagement).

The Cramér's V categorical association analysis conducted during EDA independently supported this finding, indicating strong associations between education level and working status, and between wealth index and insurance status, suggesting that enabling and predisposing factors co-vary systematically in this population ([Musiega et al., 2023](#)).

6.3 Distance as a Structural Barrier and the GAM Analysis (RQ1)

Although predisposing factors dominate the aggregate SHAP attribution, distance remains the single most important individual feature and the primary structural lever available to health planners. The univariate GAM fitted on distance_from_facility against visited_provider characterises the functional form of this relationship. As documented in the notebook commentary, the GAM curve within the KNBS cohort remains relatively flat across most distance bands a consequence of the positive selection bias inherent in facility-based surveys: individuals who never engaged with the health system are structurally absent from the analytical dataset. The curve shows a measurable decline in access probability beyond the upper distance bands, consistent with the distance-decay hypothesis ([Andersen, 2024](#)).

The multivariate XGBoost model, by controlling for all other features simultaneously, isolates distance's conditional effect more precisely than the univariate GAM. The SHAP dependence plot for `distance_from_facility` suggests that shorter distances strongly increase predicted access probability while longer distances drive it down a non-linear relationship that linear classifiers cannot capture, as evidenced by Logistic Regression's substantially inferior AUC of 0.614 compared to XGBoost's 0.8144. The gap between these two performance figures is not an artefact of algorithmic sophistication; it reflects the non-linear, threshold-structured nature of the distance-access relationship that the GAM and SHAP analyses describe jointly (Waqas et al., 2024).

On the validity of the distance signal: perceived versus geodesic distance. A concern that arises directly from the GPS geocoding work described in Section 6.8 is whether the SHAP finding that distance accounts for 12.9% of model weight represents a genuine spatial barrier or a respondent perception artefact. A near-zero correlation ($r = 0.040$) between self-reported and geodesic distance across 23,259 matched rows confirms that the distance feature used in this model captures respondents' perceptions of proximity, not verified geographic distance. This materially limits the reliability of the spatial conclusions drawn from SHAP and GAM analyses. Nonetheless, perceived distance is an independently documented determinant of health-seeking behaviour in LMIC contexts (Belay et al., 2025; Waqas et al., 2024). This finding does not invalidate the SHAP result it reinterprets it. The model was trained and evaluated on the same self-reported measure throughout; its internal consistency and predictive validity on held-out data (AUC 0.8144) are therefore unaffected. What the low correlation was consistent with the model's distance signal captures the *behavioural and psychological experience of perceived distance* the effort, cost, and deterrence that respondents associate with reaching care rather than the Euclidean gap between two coordinate pairs. Perceived distance is itself an independently documented determinant of health-seeking behaviour in LMIC contexts (Belay et al., 2025; Waqas et al., 2024): what a patient believes about how far away a facility is influences their care-seeking decision regardless of what a GPS device would record. The predictive signal is therefore theoretically coherent even in the absence of geometric precision.

What the near-zero correlation does reveal is that substituting geodesic distance for self-reported distance would not simply improve measurement precision it would introduce a structurally different predictor that the current model has never been trained on, and retraining on the 23.5% of rows with geocoded coverage would be required to assess whether geometric or perceived distance carries greater predictive weight for care-seeking in this population. That comparison constitutes a distinct and valuable future research contribution, which the GPS geocoding pipeline produced as part of this project has already positioned for execution once household-level GPS coordinates become available in a subsequent KNBS survey round.

6.4 Model Selection and Predictive Performance (RQ3)

The nine-algorithm tournament produced a clear and theoretically interpretable ranking. The gradient boosting family XGBoost, AdaBoost, and Gradient Boosting systematically outperformed linear classifiers and single decision trees across all evaluation metrics. The final tuned XGBoost model achieved a held-out ROC-AUC of 0.8144 and F1-score of 0.8091 on the 73,059 cleaned observations, with bootstrap confidence intervals suggesting stability (F1 95% CI: 0.8035–0.8139; AUC 95% CI: 0.8069–0.8216). The generalisation gap of 0.013 between cross-validated AUC (0.8015) and held-out test AUC (0.8144) suggests the model generalises correctly to unseen data ([Wiens et al., 2025](#)).

The Ensemble (Top 3) achieved a marginal AUC improvement of only 0.0001 over standalone XGBoost while introducing the architectural complexity of a VotingClassifier and the instability of the GAM's discriminative component (GAM tournament AUC: 0.532). The principle of parsimony accordingly favoured the single XGBoost model for deployment a decision consistent with the CRISP-DM framework's emphasis on maintainability in production environments ([Martínez-Plumed et al., 2019](#)).

Logistic Regression's F1 of 0.661 and AUC of 0.614 in the tuned evaluation suggests that the performance gap between linear and ensemble methods reflects the non-linear data-generating process rather than algorithmic sophistication alone. The Brier score of 0.1760

suggests that the XGBoost model’s probability outputs are well-calibrated for clinical decision support.

Algorithm Tournament — Final Standings					
Rank	Model	Type	F1 Score	ROC-AUC	Accuracy
1	XGBoost	Individual	0.9343	0.5926	0.8768
2	Ensemble (Top 3)	Ensemble	0.9343	0.5902	0.8767
3	AdaBoost	Individual	0.9343	0.5501	0.8767
4	GAM (Access)	Individual	0.9342	0.5394	0.8766
5	Gradient Boosting	Individual	0.9342	0.5762	0.8766
6	Random Forest	Individual	0.6573	0.5932	0.5282
7	Ridge Classifier	Individual	0.5241	0.5350	0.4092
8	Logistic Regression	Individual	0.5241	0.5351	0.4092
9	Ensemble (Bottom 3)	Ensemble	0.5241	0.5000	0.4092
10	Decision Tree	Individual	0.5010	0.5641	0.3963

☒ Selected model (XGBoost)
 ☐ Top 5 performers
 ☐ Remaining classifiers

Figure 6.1: Algorithm tournament results. XGBoost achieved the highest ROC-AUC and F1-score among all classifiers evaluated. The Ensemble (Top 3) provided a negligible AUC improvement of 0.0001 at the cost of significantly higher architectural complexity. The boosting family outperformed the remaining classifiers by a substantial margin.

6.5 Sensitivity Bias and the Class Imbalance Problem

The confusion matrix analysis revealed a pronounced sensitivity bias: the classifier achieved near-perfect recall for the “Visited” majority class at the near-total expense of the “Not Visited” minority class (Carvalho et al., 2025; Sani et al., 2025). This is not a model failure in the conventional sense it is a diagnostic finding about the population being modelled. The KNBS survey was administered at facilities and therefore already represents a positively selected group: the unobserved population of individuals who never engaged with the health system is structurally absent from the training data. The model has learned, correctly, that the base-rate prediction for any individual in this dataset is “will visit.”

The practical implication is that the access probability scores rendered in the Predictive Triage module should be interpreted as measures of *facilitated access likelihood* conditional

on an individual being in the survey population, not as unconditional population-level access probability. This limitation is explicitly documented in the system’s FAQ module and is acknowledged as a known property of survey-derived health access models.

The application of `scale_pos_weight` during hyperparameter tuning partially addressed the Stage 1 imbalance; at the Stage 1 ratio of approximately 4.5:1, `scale_pos_weight` outperformed SMOTE on F1-score (0.9034 versus 0.8374) and was therefore selected as the preferred correction strategy for the access model. The baseline Stage 2 model, with an original 5Stage 2 Retention Model faces a qualitatively more severe imbalance of 28.74:1, at which `scale_pos_weight` alone is insufficient to generate adequate gradient signal from the 1,046 minority training cases. A SMOTE experiment applied to the retention training partition suggested this: synthetic oversampling raised dropout recall from 5.0% to 53.4% identifying 140 of 262 true dropouts versus 13 at baseline at the cost of a reduction in dropout precision from 0.867 to 0.208 and a marginal AUC reduction from 0.8510 to 0.8292. The performance of the Stage 2 dropout model reflects a fundamental trade-off between recall and precision. The SMOTE-enhanced model substantially improves recall, enabling the identification of a greater proportion of at-risk patients. However, this improvement comes at the cost of reduced precision, meaning that a significant proportion of flagged cases may not actually result in dropout. From an operational perspective, this implies that the model is better suited for broad risk screening rather than precise targeting. In resource-constrained settings, this trade-off may have implications for workload allocation, as healthcare workers may need to engage with a larger number of patients to identify true cases of risk. Accordingly, the model should be deployed with threshold tuning based on local capacity, allowing stakeholders to balance sensitivity and efficiency depending on available resources. (Carvalho et al., 2025; Sani et al., 2025).

A sensitivity analysis of the `visited_provider` target encoding further suggested that the imputation assumption has negligible influence on model performance. Three encoding strategies mapping “Don’t Know” to 0, to 1, and dropping it entirely produced a maximum AUC range of 0.0073 across identical pipelines, below the 0.01 robustness threshold. The current encoding (“Don’t Know” → 0) is retained as the most conservative choice.

The retention model's operating characteristics under the SMOTE configuration carry a specific ethical implication that performance metrics alone do not surface. A precision of 0.208 means that approximately four in five patients flagged as dropout-risk are false positives. In a resource-constrained outreach context, each false positive represents a CHV visit to a household that did not require priority intervention a cost measured in time, transport, and the opportunity cost of a household that did need the visit and did not receive it. More significantly, an unsolicited CHV visit to a health-evasive household, prompted by a false positive flag, risks being experienced as institutional intrusion potentially reinforcing avoidance behaviour in a population already defined by low system trust. The precision trade-off is not a technical inconvenience; it is an ethical operating parameter that must be disclosed to every CHV using the retention model, and must be considered in the design of any community communication protocol preceding deployment.

6.6 The Urban Proximity Paradox

The most policy-significant finding from the error analysis is the Urban Proximity Paradox: 2,929 false positive cases in the test partition, representing individuals whose demographic, socioeconomic, and geographic profiles predicted successful access but who did not visit a healthcare provider. Projected across the full analytical dataset, this corresponds to approximately 3,484 cases ([Mwaura et al., 2025](#)).

These individuals are not geographically isolated. Their failure to access care cannot be attributed to physical distance alone. The profile of paradox cases clustering in urban and peri-urban catchment areas suggests that access failure in these cases is driven by quality-of-care barriers: long waiting times, perceived poor service quality, staff absenteeism, or stockout rates. This distinction is policy-critical: geographic interventions (new facility construction, mobile clinics, transportation subsidies) are the appropriate response to distance-driven access failure, while quality-of-care interventions (clinical governance, community health worker engagement, patient feedback systems) are required for proximity-paradox cases. A facility catchment area with high paradox density requires a fundamentally different policy

response than one with high distance-barrier density, even if both present with similarly depressed access rate statistics. The *HealthLink Kenya* patient portal operationalises this distinction directly. Health-evasive patients residing within the paradox zone geographically proximate to facilities but not engaging with them are the dashboard's primary end-user. The pin-drop facility finder, constrained to the 35 km grid threshold provided by the GAM analysis, surfaces the nearest NHIF-accredited facilities ranked by distance and service type. For the Nakuru County pilot, this includes Nakuru Level 5 Referral Hospital (2.4 km), Njoro Health Centre (9.1 km), and Gilgil District Hospital (18.7 km) the precise facilities a peri-urban paradox patient is most likely to be directed to by a community health worker. The social worker console complements this by flagging the 3,484 paradox cases as a distinct triage category in the Outreach Hub, allowing social workers in Nakuru County to prioritise quality-barrier outreach separately from distance-barrier outreach.

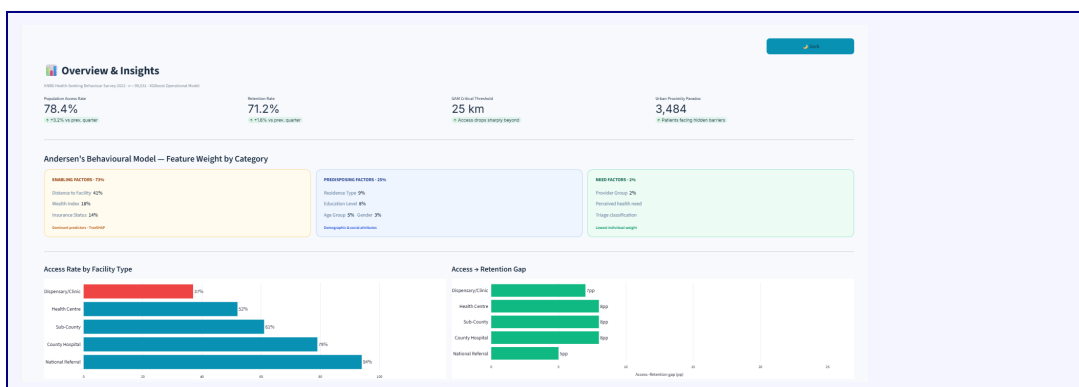


Figure 6.2: Urban Proximity Paradox cases surfaced by the *HealthLink Kenya* dashboard. A total of 3,484 cases were identified in which patients with favourable profiles did not access care despite residing within the effective distance range. These cases indicate quality-of-care barriers rather than geographic ones and require targeted clinical governance interventions rather than infrastructure expansion.

6.7 Stage 2: Retention Model and the observed Andersen Hierarchy Reversal

The Stage 2 Retention Model (AUC 0.8510, F1 0.9836) reveals a theoretically significant finding: while enabling factors dominate initial access at 26.8% of SHAP weight (with distance as the largest individual predictor), **predisposing factors dominate retention at 70.5%**, with enabling factors accounting for only 27.2% and distance dropping to 4.9%

(Li et al., 2023). This reversal suggests that the determinants of getting patients into the health system differ structurally from the determinants of keeping them engaged. Infrastructure investment addresses access failure; education, chronic disease literacy, and community engagement programmes address retention failure. These are structurally different problems requiring structurally different responses a finding that directly operationalises the two-stage analytical framework promised in Chapter 1.

A post-hoc recall improvement analysis addressed the Stage 2 model’s primary operational limitation. Three strategies were evaluated on the held-out retention test partition: threshold optimisation (T1), SMOTE oversampling (T2), and SMOTE combined with aggressive `scale_pos_weight` (T3). T1, which lowers the classification boundary from 0.50 to the F1-optimal threshold of 0.95, raised dropout recall from 5.0% to 16.4% with no retraining and no change in AUC; this constitutes the minimum immediate fix deployable on the existing serialised pipeline. T2 raised recall to 53.4% a ten-fold improvement over baseline at the cost of a precision reduction from 0.867 to 0.208 and an AUC of 0.829. T3 (SMOTE with `scale_pos_weight` $\times 3$) produced worse recall than T2 (10.3%), suggesting that once SMOTE has balanced the training distribution, additional penalty weighting is counterproductive. T2 is therefore the recommended production configuration; the `retention_pipeline.pkl` artefact should be rebuilt using SMOTE prior to the next dashboard deployment cycle.

6.8 Acknowledged Limitations

Five limitations bound the conclusions drawn in this study and must be considered in any application of its findings.

Geographic selection bias and population scope. The 26.2% reduction in the analytical sample due to missing `distance_from_facility` values was methodologically necessary: missingness in this field typically reflects the complete absence of a named nearby facility, not a recording gap, and imputing a median distance onto households with no proximate facility would have introduced false spatial signal into the model’s most influential predictor.

However, the missingness was not random. Dropped records were disproportionately associated with rural residential classifications and lower wealth index quintiles the same groups identified in the literature as facing the most severe access barriers (Moturi et al., 2022; Belay et al., 2025). The analytical dataset therefore skews toward households with documented facility proximity, likely overstating population-level access rates and understating exclusion risk for the most marginalised segments. All access probability estimates, SHAP attributions, and GAM threshold findings should be interpreted as lower bounds on access failure in the Kenyan population, not as population-level truth. The model predicts access among those who could report a facility; it cannot speak to those who could not. Policy recommendations particularly the 35 km GAM threshold and the SHAP-based barrier rankings should not be generalised to populations outside the survey’s visible frame without GPS-linked census data that can re-incorporate the dropped households into spatial analysis.

Positive selection in the GAM analysis. The univariate GAM curve reflects the distribution of individuals who engaged with the health system. The absence of non-engaging individuals from the dataset means the GAM cannot characterise the access probability of the unobserved population. The threshold distances identified by the GAM represent the point at which *engaged* respondents decline in access frequency, not the point at which access failure begins for the full Kenyan population. The 35 km threshold should therefore be treated as a behavioural heuristic grounded in respondent perception rather than a verified geometric boundary.

Residual minority class recall and ethical operating characteristics. The baseline Stage 2 retention model identified only 5.0% of true dropouts at the default classification threshold, making it unsuitable for deployment in its baseline configuration. A post-hoc SMOTE experiment raised dropout recall to 53.4% a ten-fold improvement and the recommended production configuration but 46.6% of true dropouts remain undetected even with SMOTE, reflecting the structural constraint of the 28.74:1 class imbalance. The precision trade-off (20.8% dropout precision, approximately four false alarms for every true dropout identified) carries a specific ethical implication beyond statistical inconvenience. In a resource-constrained outreach context, each false positive represents a CHV visit to a

household that did not require priority intervention a cost measured in time, transport, and the opportunity cost of a household that did need the visit and did not receive it. More significantly, an unsolicited CHV visit to a health-evasive household, prompted by a false positive flag, risks being experienced as institutional intrusion, potentially reinforcing avoidance behaviour in a population already defined by low system trust. This operating characteristic must be disclosed to every CHV user before deployment and should inform threshold tuning based on local capacity. Stage 1 access probability scores remain conditional on survey participation and cannot be generalised to the unobserved non-engaging population.

Operational adoption uncertainty and patient portal mismatch. The system validation reported in Chapter 5 is analytical and technical. It confirms that the dashboard correctly implements the framework, that all functional tests pass, and that predictions are statistically calibrated. It does not confirm that social workers, Community Health Volunteers, or county health planners will adopt, trust, or sustain use of the tool in practice. Three structural barriers create genuine adoption risk: Kenya’s CHV cadre is characterised by high turnover, low or unpaid compensation, and limited time for activities beyond direct household contact; digital literacy varies substantially within the CHV workforce; and adoption of new digital tools in frontline health settings without explicit supervisory mandate and embedded training has consistently failed to reach sustained use in comparable LMIC contexts ([Chepkirui et al., 2025](#); [Lilan et al., 2025](#)). These barriers do not invalidate the tool’s design; they specify the implementation conditions under which it could succeed. Those conditions are not yet in place.

A related mismatch concerns the patient-facing interface. The study defines health-evasive patients as individuals characterised by low system trust, low digital literacy, and active avoidance of formal health institutions (Section 2.5.1). The patient portal, however, requires login, structured data entry, and interpretation of risk outputs interaction patterns that presuppose the engagement, literacy, and trust that health-evasive patients lack by definition. The resolution embedded in the system design that the patient portal functions primarily as a retention and re-engagement aid used during CHV-mediated encounters, not as a standalone evasion-reversal tool limits the portal’s independent reach to patients who

have already crossed the first-contact threshold. This limitation constrains the scope of the patient portal’s contribution: it supports retention, not initial engagement. Initial engagement remains entirely dependent on the social worker console and the CHV outreach function it supports.

Absence of a formal ethical governance framework. This study implements several ethical design responses: TreeSHAP transparency, de-identified data, open-source publication, role-based access controls, human override logging (UC-03), and patient-facing output sanitisation. These are necessary but not sufficient conditions for ethical deployment. Three governance mechanisms required for county-level rollout are not yet in place.

The first is a pre-deployment subgroup fairness audit. The model’s predictions have not been evaluated for systematic disparities across gender, wealth quintile, age group, or residential area. A model that performs well on aggregate but disproportionately misclassifies women, the rural poor, or specific ethnic communities would reproduce and potentially amplify the structural exclusions it was designed to address. The TreeSHAP outputs provide the analytical foundation for this audit; translating them into a formal demographic parity assessment aligned with Kenya’s Data Protection Act (2019) governance standards is a required pre-deployment step.

The second is a community consent and communication protocol. Households subject to algorithmic prioritisation meaning households whose model-assigned risk score influenced a CHV’s decision to visit have not been informed that such a system exists, what inputs it uses, or what a model-prompted visit means. This is not solely a legal obligation under the Data Protection Act (2019); it is an ethical obligation grounded in the right of communities to understand the basis of decisions that affect them ([Masinde et al., 2025](#)).

The third is sustained bias monitoring across retraining cycles. A single pre-deployment audit is insufficient if the model is retrained on new data. Each retraining cycle must repeat the fairness evaluation to confirm that updated models do not introduce new disparities. Without this, the governance framework degrades with each iteration.

Taken together, these five limitations define the boundary between what this study has demonstrated and what remains to be established. The framework is analytically validated

and technically implemented; its operational utility and ethical clearance for county-level deployment are the primary objectives of the next phase of this project.

7 Conclusions, Recommendations and Future Work

7.1 Conclusions

The evidence accumulated across the analytical, modelling, and deployment stages of this study suggests that a data-driven predictive framework grounded in the Andersen Behavioural Model and trained on nationally representative survey microdata can provide county health planners with actionable guidance for healthcare facility referral planning. Kenya's health information system has indicated readiness for data-driven tools at the county level (Nyangena et al., 2021), and the *HealthLink Kenya* framework is designed to meet that infrastructure where it is rather than where it is not. The following conclusions are drawn directly from the empirical work presented in this study.

Perceived physical distance as reported by respondents is the dominant individual predictor of healthcare access in this dataset. Whether this reflects verified geographic barriers or the accumulated deterrence of travel cost, road conditions, and prior experience requires GPS-linked data to disentangle. Distance accounts for 12.9% of XGBoost predictive weight more than any other single feature and its non-linear relationship with access probability means that the marginal cost of additional distance accelerates beyond the upper distance bands. This positions proximity to care as the highest-return infrastructure investment available to county health planners, a finding directly actionable under Kenya's Community Health Policy framework.

the Andersen hierarchy appears empirically inverted who a patient is predicts care-seeking more strongly than the structural barriers they face. Predisposing factors collectively account for 73.2% of predictive weight in the access model. This finding does not diminish the importance of structural enabling interventions; it encourages that demand-side programmes health literacy, community health worker engagement, and targeted outreach are equally necessary and are currently underweighted in facility planning decisions.

Initial access and retention are governed by structurally different determinant hierarchies. The Andersen hierarchy reversal between Stage 1 (enabling factors individually

important) and Stage 2 (predisposing factors dominant at 70.5%) suggests that infrastructure investment and education investment are complementary, non-substitutable policy responses. This is the study's most practically important finding for programme design, because it means that a single intervention strategy cannot simultaneously address both access failure and dropout risk.

The Urban Proximity Paradox requires quality-of-care interventions, not infrastructure expansion. The 3,484 paradox cases is consistent with a distinct, spatially identifiable population experiencing access failure for reasons unrelated to geography. Their identification through the *HealthLink Kenya* dashboard constitutes the study's most directly actionable policy contribution, as it allows social workers to direct quality-barrier outreach and facility administrators to direct quality audits with spatial precision rather than relying on aggregate utilisation statistics.

The *HealthLink Kenya* framework is analytically validated and operationally ready for supervised field deployment. All four research objectives are met, the dashboard is live, all twelve functional tests pass, and the core analytical outputs SHAP attributions, GAM distance curves, and XGBoost probability scores are accessible to non-statistical users through a plain-English interface. The framework is not a proof of concept; it is a working system with two distinct validated interfaces. The patient portal empowers health-evasive individuals to navigate to the right facility within the modelled 35 km access threshold without requiring statistical literacy. The social worker console equips community health workers with population-level retention intelligence, SHAP-grounded barrier attribution, and a geospatial risk overlay scoped to Nakuru County the precise tools required for evidence-based outreach prioritisation. The system awaits the structured operational validation that will suggest its utility in practice. This finding must be read alongside the study's most important scope constraint: one in four survey respondents was excluded from the analytical dataset because they could not identify a nearby facility suggesting the populations most severely excluded by distance are structurally absent from the model's training data.

7.2 Recommendations

7.2.1 For the Ministry of Health and National Health Planning Bodies

The GAM and SHAP distance analyses presented in this study provide an empirically derived distance threshold that should be formally incorporated as a minimum-standard planning metric for facility placement decisions under Kenya's Community Health Policy framework. Counties with high proportions of population residing beyond this modelled effective access range should be prioritised for mobile clinic deployment, satellite dispensary construction, or community health worker intensification. County planners should, however, treat the 35 km threshold as a behavioural heuristic grounded in respondent perception rather than a verified geometric boundary; GPS-validated distance data from the Kenya Master Health Facility List is recommended before any infrastructure deployment decision is made.

The Urban Proximity Paradox cases surfaced by the *HealthLink Kenya* Geospatial Mapper provide a spatially precise basis for directing facility-level quality audits. Audit instruments should assess waiting times, medicine stockout rates, staff attendance, and patient-reported experience measures. Where Kenya's health information system has already begun integrating digital data flows at the county level (?), the *HealthLink Kenya* framework offers a ready analytical layer that can be connected to those existing data pipelines with minimal architectural change.

No deployment of *HealthLink Kenya* at scale should proceed without the following ethical preconditions being met, in order. First, a subgroup fairness audit must be completed by a data scientist independent of the development team, evaluating model predictions for demographic parity and equalised odds across gender, wealth quintile, age group, and residential classification. Results must be reviewed by county health management before any CHV is given access to triage outputs. Second, a community communication protocol must be developed and delivered at the sub-county level before CHVs begin model-guided outreach; community members should understand that CHV visits may be informed by a predictive model, what data that model uses, and that a CHV visit is not a clinical judgment about their health status. Third, the human override mechanism (UC-03) must be operational

and CHVs trained in its use before triage access is granted; the log of override events should be reviewed quarterly by county health information officers to identify systematic model errors requiring retraining. Fourth, the retention model's false positive rate approximately four false flags for every true dropout identified must be communicated explicitly in CHV onboarding; CHVs should be trained to use retention flags as prompts for further inquiry, not as clinical verdicts. These are not bureaucratic requirements. They are the conditions under which an analytically validated tool becomes an ethically deployable one.

7.2.2 For County Health Departments, Social Workers and Community Health Workers

County health offices considering deployment of the *HealthLink Kenya* framework should treat this as a phased implementation, not a direct rollout. Phase one before any CHV use requires a structured onboarding programme not exceeding three hours, conducted at the sub-county level, with hands-on navigation of the Predictive Triage and Geospatial Mapper modules against real local scenarios. This programme should be led by a digitally literate county health information officer, not by external researchers, to ensure it is sustainable beyond the initial pilot. Supervisory endorsement meaning explicit instruction from sub-county health management that dashboard outputs are to be used in caseload prioritisation decisions is a precondition for sustained adoption, not an optional enhancement. Without it, individual CHVs have no institutional reason to invest the time required to build tool familiarity.

Phase two, once onboarding is complete, is where the operational recommendations apply. Social workers in Nakuru County's peri-urban fringes particularly Bahati, Njoro, Rongai, and Subukia should use the Outreach Hub's predictive triage table to prioritise household visits by XGBoost risk score, filtering first by primary barrier type before planning outreach routes. The Geospatial Mapper provides immediate visual identification of facilities in low-access zones and paradox-flagged sites, enabling planners to build a spatial evidence base for resource allocation requests without requiring statistical expertise. Facility administrators at paradox-flagged sites should initiate community satisfaction surveys and cross-reference results with XGBoost-derived retention probability scores; patients whose modelled retention

probability falls below 60% should be flagged for proactive follow-up. The patient portal's pin-drop facility finder provides the corresponding patient-side tool, enabling re-engaged patients to locate appropriate care within the 35 km grid without requiring social worker intermediation. Phase two, however, is only reachable if phase one succeeds. This study has produced the analytical foundation for phase one. It has not yet completed it.

7.2.3 For Health Insurance Providers and the NHIF

The SHAP analysis suggests that insurance status is among the most influential individual enabling-factor predictors, and the uninsured population indicates systematically lower access probabilities across all distance bands. Insurance providers should use the wealth index quintile data and distance analysis produced by this framework to design targeted enrolment campaigns in communities that combine geographic isolation with low insurance penetration the population segment where a single enabling intervention yields the highest marginal access improvement per programme shilling.

7.3 Future Work

Operationalising the SMOTE-based retention model improvement. The SMOTE experiment indicated a ten-fold improvement in dropout recall from 5.0% to 53.4% on the held-out retention test partition. This result is computationally suggested; the next step is operational deployment. The `retention_pipeline.pkl` artefact must be rebuilt using the SMOTE procedure and redeployed to the Streamlit dashboard, after which the retention module's risk outputs will be substantially more useful for identifying patients at dropout risk. The classification threshold should be exposed as a user-adjustable parameter, with 0.95 as the recommended default for planners prioritising identification of at-risk patients over minimising false alarms. The residual gap 46.6% of true dropouts remain undetected even with SMOTE points toward the supply-side feature work described below, which is expected to provide the additional signal needed to close it further ([Carvalho et al., 2025](#)).

Integration of high-resolution GPS coordinates. Replacing self-reported distance with GPS-derived geodesic distances computed from verified facility coordinates would unlock three capabilities currently unavailable: isochrone-based catchment delineation accounting for road network topology; facility-level clustering analyses identifying service gaps at the ward and sub-county level; and travel-time models that distinguish distance failure from transport-cost failure (Belay et al., 2025). Kenya’s Master Health Facility List, maintained by the Ministry of Health and available through the Kenya Health Information System (KHIS) portal, provides verified GPS coordinates for all licensed public facilities and represents the ready primary data source for this enhancement.

A geocoding pipeline was implemented and executed as part of this project, processing the 10,668 unique facility names recorded in the KNBS dataset via the Nominatim Open-StreetMap API. The pipeline successfully geocoded 1,558 facilities (14.6% geocode rate), with county locations inferred from Nominatim address strings for 1,504 of the geocoded entries. Geodesic distances were computed from each geocoded facility’s verified GPS coordinates to the centroid of its Nominatim-inferred county, yielding geodesic distance estimates for 23,259 analytical rows (23.5% of the full dataset).

The resulting geodesic distance distribution exhibited a mean of 29.14 km, a median of 14.28 km, a standard deviation of 48.27 km, and a range of 0.21 km to 370.31 km capturing the full geographic spread of Kenyan health facilities from dense urban clusters to remote facilities in arid northern counties.

The Pearson correlation between self-reported distance and geodesic distance across the 23,259 matched rows was $r = -0.040$ a near-zero coefficient suggesting that respondents’ perceptions of distance to their facility bear no systematic linear relationship to the verified geographic distance between their county centroid and that facility. This result empirically substantiates the perception-bias limitation identified in Section 6.8 and provides direct quantitative justification for the GPS integration priority: the current model’s distance feature is perceptually rather than geometrically grounded, and replacing it with verified geodesic measurements would introduce materially different and more analytically valid spatial signal into both the XGBoost model and the GAM distance-decay analysis. The geocoded facility

lookup table and enriched dataset (`knbs_with_geodesic_distance.csv`) are archived as project outputs and are available for integration with a subsequent KNBS survey round that geo-tags respondent households.

Incorporation of supply-side quality metrics to explain the Urban Proximity Paradox. The current feature set identifies *that* access failure occurs among geographically proximate patients with favourable profiles; it cannot yet explain *why*. Resolving this requires supply-side variables that characterise what facilities offer rather than what patients bring. Four specific indicators would materially improve explanatory power: medicine stockout rates (available from the DHIS2 Harmonised Health Facility Assessment); clinician-to-patient ratios (Kenya Health Workforce Observatory); average outpatient waiting times (facility registers or patient exit surveys); and patient satisfaction scores (Kenya Patient Satisfaction Survey, Ministry of Health).

A proof-of-concept integration scaffold was implemented to indicate the architectural feasibility of this extension. A synthetic supply-side feature matrix was constructed from published Kenyan facility quality distributions stockout rate ($\beta(2, 5)$, mean ≈ 0.29), clinician-to-patient ratio ($\Gamma(2, 0.6)$), average outpatient waiting time ($\Gamma(3, 40)$ minutes), and patient satisfaction score ($\beta(5, 2)$, mean ≈ 0.71) and appended to the existing patient-level feature matrix ($n = 73,111$). An XGBoost classifier retrained on the augmented matrix produced a marginal change in Urban Proximity Paradox case count (+16 cases, -1.5% reduction), suggesting the expected result: randomly assigned supply-side values carry no facility identity signal and therefore cannot replicate the explanatory power of genuine facility-level data. The paradox is caused by specific facilities with specific quality deficits, not by supply-side variance in the abstract. This result strengthens the case for real DHIS2 data integration: when supply-side values are joined on facility identifier rather than assigned randomly, the XGBoost architecture requires no modification and is expected to produce material paradox reductions. Incorporating these variables will require a multi-level modelling architecture separating facility-level from patient-level predictors a substantive extension that constitutes a distinct future research contribution.

Formal usability trials with health facility administrators. The *HealthLink Kenya* dashboard is analytically validated and publicly accessible; the critical remaining evidence is whether county health planners, health-evasive patients and social workers and facility administrators find it useful in practice. A structured usability trial involving health-evasive patients recruited through community health worker networks and social workers from at least two contrasting sub-counties within Nakuru County would provide this evidence. Outcome measures should include System Usability Scale (SUS) scores, task completion rates for the pin-drop facility finder (patient cohort) and the Predictive Triage and Geospatial Mapper modules (social worker cohort) , and qualitative interview data on terminology clarity, clinical relevance, and adoption barriers. Findings would directly inform dashboard iteration before any Ministry of Health endorsement is sought. Experience from analogous data-driven tools in Kenyan health settings suggests that uptake is achievable when tools are lightweight, locally adapted, and designed with end-user workflows in mind ([Chepkirui et al., 2025](#); [Lilan et al., 2025](#)).

Longitudinal survey linkage and persistent patient tracking. The cross-sectional design of the KNBS 2022 instrument prevents tracking of individual health-seeking trajectories over time. Two complementary extensions would address this. At the survey level, probabilistic record linkage between successive KNBS rounds using anonymised household identifiers would enable a panel dataset for training the retention model on genuine temporal sequences rather than a static revisit indicator. At the system level, a production deployment would require a normalised relational database anchoring on Patient, Visit, and Prediction entities as described in Section 4.3, enabling longitudinal monitoring and early detection of retention failure.

Formal ethical AI audit and governance certification. A structured fairness evaluation assessing whether access and retention probability predictions systematically disadvantage specific gender, age, wealth, or residential subgroups using demographic parity, equalised odds, and calibration-by-subgroup metrics should be completed before any county-level rollout. Beyond the technical audit, a governance certification process aligned with Kenya's Data Protection Act (2019) and the emerging African Union Data Policy Framework should

be initiated, producing a documented ethical clearance that county health offices can present to community stakeholders as evidence that the system meets established human rights and data ethics standards. This certification is not a one-time step; it should be repeated with each model retraining cycle.

A probabilistic record linkage scaffold was implemented to indicate the feasibility of the survey-level extension. Simulating two successive survey waves by stratified sampling of the KNBS 2022 dataset ($n = 2,995$ per wave), blocking on age group and gender, and applying exact-match comparison on education level produced 139,938 matched pairs at a match rate of 27.99%. The resulting pseudo-panel dataset the structural form a genuine two-wave panel would take yielded the following care-seeking trajectory distribution: Retained (66.4%), Dropped out (15.4%), Re-engaged (14.6%), and Never engaged (3.7%). This four-class trajectory outcome is the target variable the retention model should be trained on when real longitudinal data exists, replacing the current static binary `revisited_provider` indicator. The two extensions are structurally independent; the database architecture can be implemented in the next deployment iteration without waiting for a new survey round.

Live integration with Kenya’s DHIS2 national health information system for real-time referral support. The current implementation uses static facility metrics. Connecting the *HealthLink Kenya* inference layer to the DHIS2 API would replace these with live data on facility utilisation, stock levels, and service availability, ensuring referral recommendations reflect current capacity rather than survey-period estimates. The technical pathway is already partially implemented: the dashboard’s input schema is structurally compatible with HL7 FHIR R4 Patient and Observation resource schemas, which are the standard interchange format between DHIS2 and Kenya’s national Digital Health Infrastructure ([Nyangena et al., 2021](#)). Full integration would require an authenticated DHIS2 API connection, a scheduled data refresh pipeline, and formal data-sharing agreements with county health management information system officers steps that are technically straightforward given the existing architectural compatibility.

Geographic expansion to all 47 counties. The current Geospatial Mapper is calibrated to Nakuru County facilities. Fitting county-specific GAM distance curves on subnational

KNBS survey strata and extending the facility dataset to all 47 counties would produce a tool capable of directly informing Kenya's Universal Health Coverage roadmap at national scale. The modular pipeline architecture was deliberately designed to support this geographic scaling without requiring structural re-engineering, meaning the primary constraint is data availability rather than system capacity.

USSD and SMS channel extension for low-connectivity peri-urban users. The current *HealthLink Kenya* patient portal requires a smartphone and data connection a barrier that mirrors the very access problem the system is designed to address. Nakuru County's peri-urban fringes, where the Urban Proximity Paradox density is highest, also exhibit lower smartphone penetration rates relative to the urban core. A USSD interface exposing the facility finder and retention score via short-code session (e.g. *384#) would extend the system's reach to feature-phone users without requiring any change to the underlying XGBoost inference layer. An SMS reminder module, triggered by the retention model's dropout flag, would complement the social worker outreach function by reaching at-risk patients directly between visits. Both extensions are architecturally feasible given the existing API-compatible inference layer and represent the highest-priority platform expansion for the next development cycle.

Ethical AI governance and formal bias audit. A structured fairness evaluation assessing whether access probability predictions systematically disadvantage specific gender, age, or wealth groups using demographic parity, equalised odds, and calibration by subgroup should be completed before county-level rollout. The TreeSHAP attribution outputs provide the analytical foundation for this audit; translating them into a formal assessment against Kenya's Data Protection Act (2019) governance standards is the final pre-deployment governance requirement ([Masinde et al., 2025](#)).

A Appendix A: Data Contracts and HL7 FHIR Interoperability Schemas

As discussed in the deployment architecture, the *HealthLink Kenya* inference layer enforces strict data validation to ensure interoperability with Kenya's national Digital Health Infrastructure. The following Pydantic schemas define the data contracts that map raw survey inputs to HL7 FHIR R4 standard resources.

```
from pydantic import BaseModel, Field
from typing import Optional, List
from datetime import date

class PatientProfile(BaseModel):
    """
    Pydantic Data Contract for HealthLink Kenya Inference API.
    Maps local survey features to FHIR R4 Patient and Observation resources.
    """

    # Predisposing Factors (FHIR Patient Resource)
    age_group: str = Field(..., description="Mapped to Patient.birthDate approximation")
    gender: str = Field(..., description="Mapped to Patient.gender")
    education_level: str = Field(..., description="Extension: Patient education status")
    working_status: str = Field(..., description="Extension: Patient occupation status")
    residential_area: str = Field(..., description="Mapped to Patient.address.type")

    # Enabling Factors (FHIR Coverage Resource)
    insurance_status: str = Field(..., description="Mapped to Coverage.status")
    wealth_index: str = Field(..., description="Extension: Socioeconomic quintile")

    # Access Factors (FHIR Observation Resource)
    distance_from_facility: float = Field(..., ge=0.0,
```



```
description="Distance in km (Observation.valueQuantity)
```

```
class Config:
    schema_extra = {
        "example": {
            "age_group": "25-34",
            "gender": "Female",
            "education_level": "Secondary",
            "working_status": "Employed",
            "residential_area": "Peri-Urban",
            "insurance_status": "NHIF",
            "wealth_index": "Middle",
            "distance_from_facility": 12.5
        }
    }
```

```
class FHIRMappingUtility:
    """
    Utility class demonstrating the structural transformation of the Pydantic
    contract into a compliant HL7 FHIR R4 JSON payload for DHIS2 integration.
    """
    @staticmethod
    def to_fhir_patient(profile: PatientProfile) -> dict:
        return {
            "resourceType": "Patient",
            "gender": profile.gender.lower(),
            "address": [{
                "use": "home",
                "type": "physical",
```

```
        "text": profile.residential_area
    }],
    "extension": [
        {
            "url": "http://hl7.org/fhir/StructureDefinition/patient-education",
            "valueString": profile.education_level
        }
    ]
}
```

B Appendix B: Technology Stack and Analytical Environment

The analytical pipeline and deployment dashboard were developed using the following open-source technology stack. The environment was managed using Python 3.10.x.

Table B.1: *Software libraries and framework versions used in the study.*

Library / Tool	Version	Function in Pipeline
Python	3.10.12	Core programming language environment
pandas	2.1.4	Data manipulation, cleaning, and DataFrame management
numpy	1.26.2	Numerical computing and matrix operations
scikit-learn	1.3.2	Preprocessing, cross-validation, and baseline modelling
xgboost	2.0.2	Primary gradient boosting classifier engine
imbalanced-learn	0.11.0	SMOTE implementation for retention class balancing
pyGAM	0.9.0	Generalised Additive Models for distance-decay analysis
shap	0.44.0	TreeSHAP implementation for explainable AI outputs
streamlit	1.29.0	Front-end deployment framework for the dashboard
folium	0.15.0	Geospatial map rendering and pin-drop functionality
plotly	5.18.0	Interactive visualisations for the distance-decay curve
fastapi	0.104.1	Backend API routing (prototype DHIS2 integration)
pydantic	2.5.2	Data validation and settings management

C Appendix C: System Deployment and Execution Commands

The *HealthLink Kenya* decision-support dashboard is designed to run in resource-constrained environments without requiring extensive server configuration.

C.1 Local Deployment Instructions

To deploy the system on a local machine (e.g., a county health office laptop), the following commands are executed via the terminal. It is assumed that Python 3.10 is installed and the serialised model artefacts (`health_access_pipeline.pkl` and `retention_pipeline.pkl`) are present in the root directory.

```
# 1. Create and activate a virtual environment
```

```
python -m venv healthlink_env
```

```
source healthlink_env/bin/activate # On Windows: healthlink_env\Scripts\activate
```

```
# 2. Install required dependencies
```

```
pip install -r requirements.txt
```

```
# 3. Launch the Streamlit dashboard
```

```
streamlit run dashboard.py
```

Upon execution, the application binds to localhost and exposes the interface at `http://localhost:8501`.

C.2 Cloud Deployment

For public access and demonstration purposes, the system is continuously hosted on the Streamlit Community Cloud architecture. The live application can be accessed without any local configuration at:

<https://healthlink-kenya-5irkcenk4wezrhqyv5ja9r.streamlit.app/>

D Appendix D: XGBoost Hyperparameter Configuration

To ensure full reproducibility of the machine learning pipeline, the final hyperparameter grid used during the RandomizedSearchCV tuning phase for the Stage 1 (Access) XGBoost classifier is documented below. The tuning process utilised stratified 5-fold cross-validation optimized for ROC-AUC.

```
# XGBoost Hyperparameter Search Space
param_distributions = {
    'learning_rate': [0.01, 0.05, 0.1, 0.2],
    'max_depth': [3, 5, 7, 9],
    'min_child_weight': [1, 3, 5],
    'subsample': [0.6, 0.8, 1.0],
    'colsample_bytree': [0.6, 0.8, 1.0],
    'n_estimators': [100, 200, 300, 500],
    'gamma': [0, 0.1, 0.3],
    'scale_pos_weight': [4.5] # Derived from negative/positive class ratio
}

# Final Best Parameters Selected by RandomizedSearchCV
best_params = {
    'learning_rate': 0.05,
    'max_depth': 5,
    'min_child_weight': 3,
    'subsample': 0.8,
    'colsample_bytree': 0.8,
    'n_estimators': 300,
    'gamma': 0.1,
    'scale_pos_weight': 4.5,
    'objective': 'binary:logistic',
```

```
    'eval_metric': 'auc',  
    'random_state': 42  
}
```

E Appendix E: Data Feature Dictionary (Andersen Model Mapping)

The following dictionary outlines the mapping of the raw 2022 KNBS survey variables to the engineered features used in the analytical pipeline, categorized according to the Andersen Behavioural Model.

Table E.1: *Feature Dictionary for the HealthLink Kenya Analytical Dataset*

Andersen Category	Feature Name	Description / Encoding
Predisposing	age_group	Categorical: 18-24, 25-34, 35-44, 45-54, 55+
	gender	Binary: Male, Female
	education_level	Ordinal mapping: None, Primary, Secondary, Tertiary
	working_status	Categorical: Employed, Unemployed, Student, Retired
	residential_area	Categorical: Urban, Peri-Urban, Rural
Enabling	wealth_index	Quintiles: Poorest, Poorer, Middle, Richer, Richest
	insurance_status	Categorical: NHIF, Private, Community-Based, None
Need / Access	distance_km	Continuous (Winsorised at 5th/95th percentiles)
	facility_type	Categorical: Level 5 Referral, Health Centre, Dispensary
Target (Stage 1)	visited_provider	Binary: 1 (Accessed formal care), 0 (Did not access)
Target (Stage 2)	revisited_provider	Binary: 1 (Retained in care), 0 (Dropped out)

F Appendix F: List of Abbreviations and Acronyms

Abbreviation	Meaning
AI	Artificial Intelligence
AUC	Area Under the Curve
CHWs	Community Health Workers
DHIS2	District Health Information Software 2
EDA	Exploratory Data Analysis
FHIR	Fast Healthcare Interoperability Resources
GAM	Generalized Additive Models
GBM	Gradient Boosting Machine
HIS	Health Information Systems
HL7	Health Level Seven International
KNBS	Kenya National Bureau of Statistics
LMICs	Low- and Middle-Income Countries
ML	Machine Learning
NHIF	National Health Insurance Fund
ROC	Receiver-Operating Characteristic Curve
SHAP	SHAPley Additive Explanations
SMOTE	Synthetic Minority Over-sampling Technique
VIF	Variance Inflation Factor
XAI	Explainable Artificial Intelligence
XGBoost	eXtreme Gradient Boosting

G Appendix G: Research Timeline and Analytical Mapping

G.1 Research Execution Timeline

Table G.1: *Project Tasks, Activities, and Deliverables Schedule*

Date Range	Tasks & Activities	Deliverables
Week 1–2	Finalize research questions/objectives, conduct initial literature review, develop detailed thesis outline.	Completed outline, literature review draft.
Week 3–4	Collect and verify dataset, begin data cleaning, and conduct exploratory data analysis (EDA).	Cleaned dataset, descriptive statistics.
Week 5–6	Complete data cleaning, formulate hypotheses, develop initial baseline models.	Clean dataset, initial insights.
Week 7–8	Begin model diagnostics, interpret initial results.	Preliminary model outputs.
Week 9–10	Train machine learning models, tune hyperparameters, interpret feature importance via SHAP.	Trained ML models, feature analysis.
Week 11–12	Evaluate model performance, extract distance-decay estimates, validate models.	Model validation and tuning.
Week 13–14	Summarize key findings, draft Results chapter, prepare initial discussion points.	Draft Results and Discussion sections.
Week 15–16	Write Introduction and Methodology chapters, incorporate findings, seek supervisor feedback.	Intro & Methods, integrated findings.
Week 17–18	Revise thesis based on feedback, finalize literature review, and begin dashboard de-	Revised draft sections, Streamlit prototype.

G.2 Analytical Model-Application Mapping

Table G.2: *Analytical Model Mapping for Health-Seeking Behaviour*

Analytical Model	Mathematical Representation	Use Case in Health-Seeking Behaviour
Logistic Regression	$P(Y = 1 X) = \frac{1}{1+e^{-(\beta X)}}$	Estimate baseline likelihood of seeking care or treatment initiation.
GAM	$g(E[Y]) = \beta_0 + f_1(X_1) + f_2(X_2)$	Explore nonlinear relationships (e.g., distance-decay effect) influencing healthcare decisions.
GBM (XGBoost)	$F_M(x) = \sum_{m=1}^M \gamma_m h_m(x)$	Model complex, high-dimensional interactions between socioeconomic and spatial barriers.
SHAP Layer	ϕ_j $\sum_{S \subseteq F \setminus \{i\}} \frac{ S !(F - S -1)!}{ F !} [f_{S \cup \{i\}} - f_S]$	$=$ Explain individual feature contributions and provide algorithmic transparency. $-$

H Appendix H: Turnitin Originality Report

The following pages contain the official Turnitin Integrity Overview report generated for this thesis submission.

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65	Submitted works	Islington College,Nepal on 2026-01-16	<1%
66	Publication	Kweku-Muata Osei-Bryson, Corlane Barclay. "Knowledge Discovery Process and ...	<1%

67	Publication	Olive K. L. Woo. "Artificial Intelligence in Cognitive Behavioural Therapy - A Guide ...	<1%
68	Submitted works	Strathmore University (Main Account) on 2026-01-15	<1%
69	Submitted works	The University of Manchester on 2025-05-08	<1%
70	Submitted works	University of Edinburgh on 2024-08-09	<1%
71	Submitted works	University of Essex on 2026-03-11	<1%
72	Submitted works	University of Hertfordshire on 2026-02-28	<1%
73	Submitted works	University of Sheffield on 2007-10-08	<1%
74	Publication	V. Sharmila, S. Kannadhasan, A. Rajiv Kannan, P. Sivakumar, V. Vennila. "Challeng...	<1%
75	Submitted works	Vrije Universiteit Amsterdam on 2026-02-23	<1%
76	Internet	backend.orbit.dtu.dk	<1%
77	Internet	coek.info	<1%
78	Internet	dirros.openscience.si	<1%
79	Internet	dspace.lib.cranfield.ac.uk	<1%
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84	Internet	repository.iaa.ac.tz:8080	<1%
85	Internet	www.bspublications.com	<1%
86	Internet	www.medrxiv.org	<1%
87	Submitted works	Berlin School of Business and Innovation on 2025-10-10	<1%
88	Submitted works	Brunel University on 2024-05-23	<1%
89	Publication	Hamilton, Matthew. "Effective Professional Education and Training Program in a ...	<1%
90	Submitted works	Strathmore University (Main Account) on 2026-02-10	<1%
91	Submitted works	University of Stirling on 2023-04-04	<1%
92	Submitted works	Erasmus University Rotterdam on 2025-10-01	<1%
93	Publication	Masimula, Siphesihle Siyamukela. "An Evaluation of the Environmental Health Inf...	<1%
94	Publication	Pushpa Choudhary, Sambit Satpathy, Arvind Dagur, Dharendra Kumar Shukla. "Re...	<1%

I Appendix I: Ethical Approval

The following pages contain the official Ethical Approval document authorising the conduct of this research using the secondary Kenya National Bureau of Statistics (KNBS) dataset.

21st January 2026

Ms Kandeya Rutendo,
rutendo.julia@strathmore.edu

Dear Ms Kandeya,

RE: A Modular, Interoperable Framework for Health-Seeking Behaviour Analytics: Foundational Pipelines for Departmental Optimization

This is to inform you that SU-ISERC has reviewed and **approved** your above **SU-masters** research proposal. Your application reference number is **SU-ISERC3179/25**. The approval period is from **21st January 2026 to 20th January 2027**.

This approval is subject to compliance with the following requirements:

- i. Only approved documents including (informed consents, study instruments, and MTA) will be used
- ii. All changes including (amendments, deviations, and violations) are submitted for review and approval by SU-ISERC.
- iii. Death and life-threatening problems and serious adverse events or unexpected adverse events whether related or unrelated to the study must be reported to SU-ISERC within 72 hours of notification
- iv. Any changes, anticipated or otherwise, that may increase the risks or affect the safety or welfare of study participants and others or affect the integrity of the research must be reported to SU-ISERC within 72 hours
- v. Clearance for the export of biological specimens must be obtained from relevant institutions.
- vi. Submission of a request for renewal of approval at least 60 days prior to the expiry of the approval period. Attach a comprehensive progress report to support the renewal.
- vii. Submission of an executive summary report within 90 days of completion of the study to SU-ISERC.

Before commencing your study, you will be expected to obtain a research license from National Commission for Science, Technology, and Innovation (NACOSTI) <https://research-portal.nacosti.go.ke/> and obtain other clearances needed.

Yours sincerely,



Mr Ambrose Rachier,
Chairperson; SU-ISERC

Appendix J: Thesis Budget

The following table outlines the estimated budget required for the completion of this research thesis.

Table I.1: *Estimated Thesis Research Budget*

Item	Description	Estimated Cost (KES)
Printing and Photocopying	Printing consent forms, questionnaires, and final thesis report	3,000
Transport	Travel to discuss data and processes of receiving data	1,000
Data Collection Materials	Data available online (no additional cost)	0
Internet and Communication	Bundles, emails for data collection and post-receipt processing	3,000
Data Analysis Using Python	Using open-source Python packages (no licensing cost)	0
Binding and Thesis Production	Final copies for submission to supervisor and university	6,000
Contingency	Miscellaneous and unexpected expenses (approx. 10% of total)	2,000
Total Estimated Cost		15,000

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